

TOPICAL REVIEW

Machine Learning Applications in Energy Harvesting Internet of Things Networks: A Review

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ABSTRACT The growth of Internet of Things (IoT) devices continues to experience an exponential rise due to their vast applications in various industries. However, sustaining their operational lifetime is a major challenge due to critical factors, such as the need for frequent recharging of energy buffers. The recent convergence of green energy harvesting (EH) and IoT technologies has proven to be a potential solution to this challenge. However, the intermittent characteristics of green energy sources and wireless fading channels pose another challenge, as the quality of service in IoT networks revolves around the available energy and channel conditions. The traditional optimization strategies based on non-causal knowledge about these random quantities are deemed unsuitable for realizing an autonomous IoT network operation. To combat this challenge, various algorithms from the field of machine learning (ML) have been proposed as potential solutions. Therefore, in this article, we aim to investigate the applications of ML algorithms in EH IoT networks. To achieve this, first, we provide an overview of ML categories commonly adopted in IoT networks. Secondly, due to the peculiarity of EH IoT networks, we provide an extensive description of the ML categories widely explored in this domain. This includes reinforcement learning, deep learning, and deep reinforcement learning. Thirdly, we present a review of studies where the applications of the aforementioned ML algorithms are demonstrated. Further, we identify challenges that are likely to impact the implementation of these algorithms. In conclusion, we highlight unexplored and emerging areas for potential future research.

INDEX TERMS Deep learning, deep reinforcement learning, energy harvesting, Internet of Things, machine learning, reinforcement learning.

I. INTRODUCTION

The advent of fifth-generation (5G) and beyond 5G systems, coupled with the widespread use of internet-based applications, has revolutionized the field of wireless communications. This is more evident in the Internet of Things (IoT) networks, where billions of low-power devices are deployed for various applications such as industrial and home automation, intelligent transportation systems, health and environmental monitoring, and so on [1], [2]. These devices rely on batteries with limited capacity as a power source, thus, raising concerns for the realization of long-term operational lifetime [3]. Moreover, the devices may be installed in harsh

environments or in locations difficult to access. Further, for a large-scale IoT network, frequent replacement of batteries of billions of IoT devices may become expensive or impractical, not to mention the serious environmental issues associated with improper disposal of batteries [4].

To address the aforementioned challenges, energy harvesting (EH) technology has been identified as a viable and eco-friendly solution. In this regard, EH IoTs are equipped with energy harvesters and buffers to harness and store the energy from environmental sources such as radio frequency (RF), solar, wind, triboelectric nanogenerator (TENG), etc [5], [6], [7], [8], [9]. An illustration of this scenario is depicted in Fig. 1. Although EH technology supports the autonomous operation of IoT networks and eliminates the need for traditional power supply and battery

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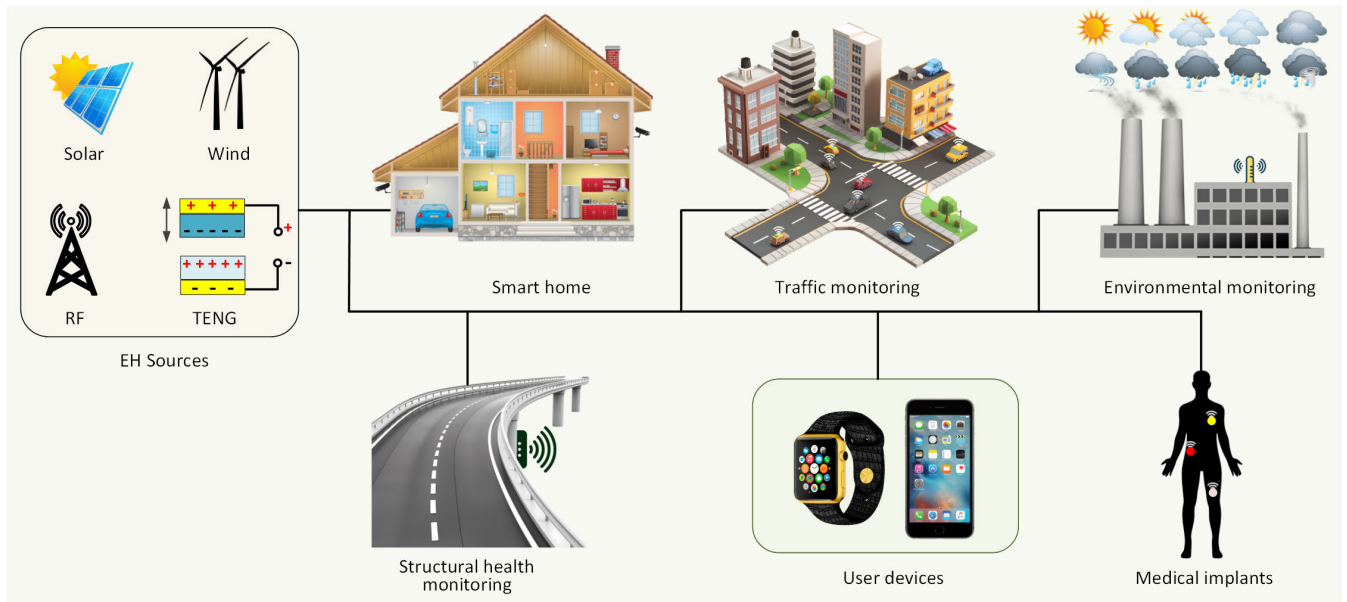


FIGURE 1. EH IoT application scenarios.

replacement, the stochastic nature of the energy sources and the wireless fading channel pose another challenge. These include the ambiguity surrounding the energy accessible for data transmission and how to efficiently allocate the harvested energy to optimize a system design objective.

The traditional EH-based solutions rely on perfect non-causal knowledge of the energy arrival rate from the source and propagation channel state [10]. In this case, the expected amount of harvested energy and the channel coefficients are predetermined. Hence, the EH node may decide to halt its transmission when the channel condition is poor or to conserve the harvested energy. Though this approach may provide a theoretical upper bound for performance analysis, relying on such a solution is impractical because EH processes and channel fading are random processes.

Interestingly, the application of various machine learning (ML) algorithms, e.g., reinforcement learning (RL) and deep learning (DL) algorithms, in EH IoT networks has become a vital tool for tackling the impracticality associated with non-causal or statistical-based EH solutions. While some of these algorithms are suitable for small-scale networks, their computational convergence may be infeasible under a large-scale network scenario due to the curse of dimensionality associated with large state-action space [11]. Moreover, the application of these ML algorithms cannot be generalized for solving different network object functions. To address the aforementioned concerns, this paper aims to provide a state-of-the-art review of various studies devoted to ML-based solutions in the EH IoT networks.

A. RELATED WORKS

In this section, we present a review of existing and related survey articles based on ML applications in EH wireless communication networks. This direction has been highlighted

in a few papers [12], [13], [14], [15], [16], [17], [18], [19], as summarized in Table 1. In [12], the application of reinforcement learning for addressing different optimization problems such as power allocation, data offloading, access control, etc., under different network contexts is discussed. Further, various RL techniques are employed, and the implementation strategies are identified. In [13], the paper focuses on intermittent deep learning-based solutions in EH IoT networks to improve inference efficiency. The study is motivated by the intermittent energy supply and high computational time and energy overhead incurred during the check-pointing of feature maps generated in the course of the deep neural network (DNN) process. Aside from the DNN-based check-pointing techniques presented, the elements of the neural network and hardware architecture in the context of intermittent inference are highlighted. In the latter section of the article, design challenges such as minimizing check-pointing overhead, developing an adaptive inference workload to improve the accuracy of the system analysis, etc., are presented. The authors of [14], discuss AI-based solutions applicable to energy and data security. In the article, solutions based on federated learning and deep learning are presented. Further, foreseeable challenges in emerging technologies, energy assessment analysis, impact of interference, beamforming optimization, and the development of energy-efficient algorithms are highlighted. In [15], a comprehensive report of thermoelectric generator (TEG) applications in ML-aided IoT is presented. In the study, the design and implementation of TEGs with emphasis on operating principles, fabrication materials, and topologies are discussed. Further, the report highlights various energy management strategies and emerging applications under different energy sources such as environmental, biological, and technological power sources. In [16], the authors highlight

four main categories in which the application processes of ML algorithms for TENG-aided sensing systems, can be classified. This includes the problem definition phase, where the problem to be solved is first analyzed, followed by the selection of a suitable ML technique. The second phase encompasses the training data acquisition, while in the third phase, different ML algorithms for modelling and training purposes are discussed. Lastly, the evaluation indices to analyze the performance of the identified algorithms are highlighted. Further to the aforementioned application processes, Zhang et al. [17], highlight different classification and predictive functions of ML algorithms in TENG-based applications. The study highlights the advantage of TENG for feature extraction due to its distinctive time series signals generated when in contact with different objects. The paper exemplifies this concept via different object recognition and pattern recognition in human activities. Further, various reports that leverage the historical data for prediction purposes are discussed. While the report in [18] adopt similar application processes described in [16] and [17], additional evaluation methods are illustrated. This includes the K-fold cross-validation, repeated learning test cross-validation, leave-one-out method, and self-help sampling method. However, reported studies indicate that these methods have trade-offs in terms of cost, complexity, and data size required for optimal performance. Unlike [16], [17], and [18], the authors of [19], extended their report to cover applications of TENG with unsupervised deep learning. Further, an in-depth discussion of various fields of TENG applications, such as fault detection, healthcare and monitoring, and human–computer interaction, is presented. The article is concluded with several challenges, such as the bio-friendliness of TENG and computational error in the ML algorithm. Moreover, potential solutions to the identified challenges are provided.

B. MOTIVATION AND CONTRIBUTIONS

The integration of ML algorithms in EH IoT networks has become a trending topic in the field of wireless communication networks. This is due to the capacity of these algorithms to deal with the energy uncertainty associated with intermittent energy sources and the random fading of wireless channels. However, due to the peculiarities of ML algorithms, their applications cannot be generalized for different objective functions in IoT network design. Though some review articles have reported various studies on the applications of ML algorithms in the EH IoT networks as presented in Section I-A and Table 1, there are several critical areas that are yet to be captured in their reports. For instance, the paper [13] is centered only on DNN to improve computational efficiency without considering the performance of other key ML techniques employed in EH IoT networks, such as RL and deep reinforcement learning (DRL). The report presented in [14] is limited to ML techniques used for the realization of energy and data security. Though these objectives are critical to efficient and secure system design, the adequacy

of the literature considered is deemed insufficient to capture other salient challenges in secure EH IoT networks, such as the identification of transmission patterns of malicious devices. The report presented in [15], is conducted under thermoelectric generator (TEG) powered networks. However, TEGs performance is generally constrained due to their low energy efficiency and high maintenance cost. Moreover, a constant heat source is required to sustain their operation, which is a major concern for the realization of smaller devices used in IoT networks. For instance, aside from safety issues, a cooling unit may be required to regulate the heat conversion process, which may increase the complexity of the hardware design. Hence, due to the aforementioned concerns, TEG-based solutions may not support the scalability of IoT networks compared to types of energy sources such as solar or RF. Another common limitation of the publications noted above is the absence of detailed information regarding the types of ML algorithms presented, which could pose a comprehension challenge for new researchers in this field. For instance, the behaviour of different ML algorithms in terms of computational complexity, training convergence rate, and approaches to handling challenges such as over-fitting, overestimation bias, and so on, are not captured. While some of these concerns are highlighted in [16], [17], [18], and [19], the studies do not cover the applications of ML algorithms at the physical transmission layer of the system design. Moreover, reports on applications of RL and DRL, which are critical to the development of intelligent and autonomous system design, are not provided. Thus, in the following, we present our contributions to the current article envisaged to bridge the identified gaps in the literature.

- To ensure appropriate background information from a technical standpoint, we provide a brief overview of various ML categories employed in wireless communication systems and later present detailed descriptions of the ones applicable to EH IoT networks. These include RL, DL, and DRL techniques.
- Based on the identified ML techniques, we provide a critical review of various convincing solutions from the literature to elucidate the current state of research in this area. In this regard, we identify the specific type of network application in which the investigation is conducted, the learning algorithm adopted, the achievements, and perceived limitations.
- We highlight several challenges that are crucial to the choice, design, and implementation of ML algorithms discussed. Lastly, we offer a variety of perspectives on fascinating topics that remain unexplored for further study.

The remainder of the paper is organized as follows: We present a concise overview of ML techniques in Section II. A comprehensive review of RL, DL, and DRL-based solutions is presented in Section III. Lessons learned, design challenges, open areas for future research, and current research trends are provided in Sections IV, V, VI, and VII, respectively. Conclusions are drawn in Section VIII. The list

TABLE 1. Summary of related review articles on ML applications in EH IoT networks.

Reference	Year	Focused ML Techniques							Remarks
		SL	UL	RL	DSL	DUL	HDL	DRL	
[12]	2019	✗	✗	✓	✗	✗	✗	✗	Covers key RL algorithms and implementation strategies employed for various optimization purposes such as power management, user scheduling data allocation etc.
[13]	2022	✗	✗	✗	✓	✗	✗	✗	Focuses on intermittent deep learning-based solutions to improve inference efficiency. Specifically, the DNN-based check-pointing technique, and key elements of neural networks and hardware architectures for intermittent inference are presented.
[14]	2023	✗	✗	✗	✓	✗	✗	✗	Discusses AI-based solutions applicable to energy and data security. Moreover, foreseeable challenges in reflective surface-based systems, interference and beamforming optimization, and the development of energy-efficient algorithms are highlighted.
[15]	2023	✗	✗	✗	✓	✗	✗	✗	Covers EH topologies, principles and materials employed for TEG fabrication. Further, the report highlights energy management strategies and emerging applications under different energy sources such as environmental, biological, and technological power sources.
[16]	2023	✓	✓	✗	✓	✗	✗	✗	Discusses the application processes of ML algorithms for TENG-aided sensing systems. This includes the problem definition and selection of suitable ML algorithms, training data acquisition, system modeling and training strategy, and evaluation indices.
[17]	2024	✓	✓	✗	✓	✗	✗	✗	Highlight the feature-extracting properties of TENG and various ML algorithms employed for processing the extracted features. Further, the paper illustrates several classification and predictive functions of ML-aided TENG.
[18]	2024	✓	✗	✗	✓	✗	✗	✗	Discusses the application processes in ML-aided TENG operation with more emphasis on various validation methods. Several challenges, such as data quality, hardware size, limitations in terms of signal processing, etc.
[19]	2024	✓	✓	✗	✓	✓	✗	✗	Highlight different working modes of TENG, followed by a comprehensive report that illustrates the integration of ML algorithms with TENG. Various fields of TENG applications, challenges, and potential solutions are presented in the latter part of the article.
This paper	2024	✓	✓	✓	✓	✓	✓	✓	Provides a concise overview of ML techniques employed in wireless communication systems. Further, a critical review of ML solutions widely adopted in EH-IoT, i.e., RL, DL, and DRL is presented. In addition, design and implementation challenges and open areas for future research are identified.

SL - Supervised learning, UL - Unsupervised learning, RL - Reinforcement learning, DL - Deep learning, DSL - Deep supervised learning, DUL - Deep supervised learning, HDL - Hybrid supervised and unsupervised deep learning, DRL - Deep reinforcement learning.

and definition of frequently used acronyms are provided in Table 2.

II. A BRIEF OVERVIEW OF ML TECHNIQUES

In this section, we aim to familiarize the reader with different categories of ML techniques adopted in IoT networks by concisely highlighting their fundamental features. Note

that we do not intend to provide an in-depth analysis of different types of ML algorithms in this section because comprehensive reports on this have already been presented in [20], [21], [22], and [23]. Nonetheless, details on the ML techniques applicable to EH IoT networks will be provided in Section III. Fig. 2 depicts the taxonomy of ML techniques adopted in this paper.

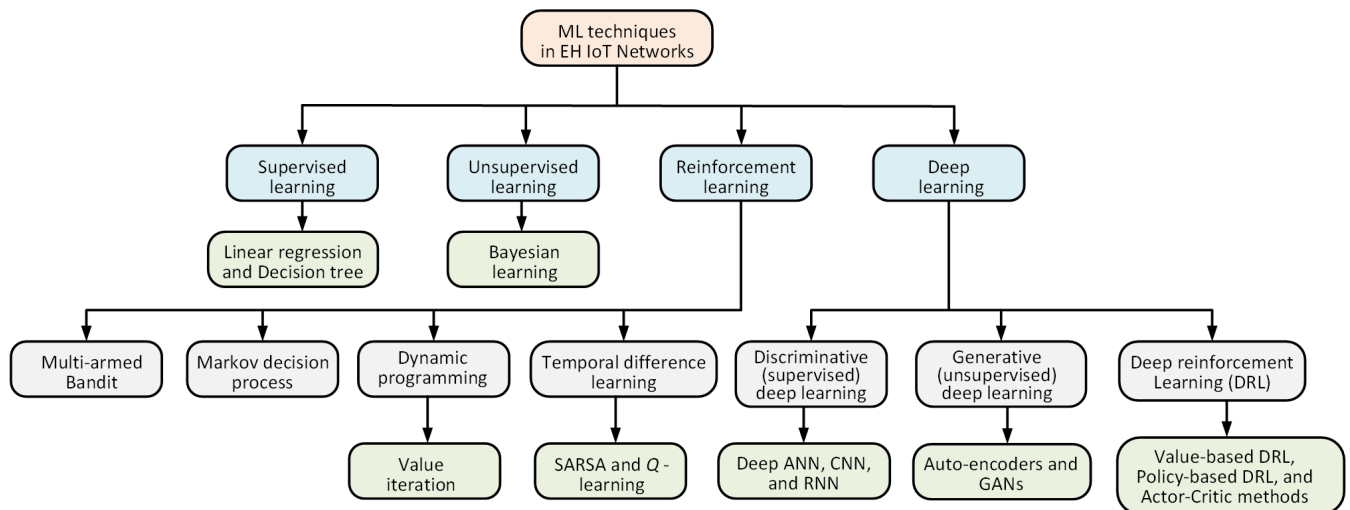


FIGURE 2. Taxonomy of ML techniques in EH IoT networks.

Using the conventional classification approach, ML algorithms can be grouped into supervised, unsupervised, and reinforcement learning [24].

- **Supervised learning:** In this learning method, labeled training data samples, known a priori, are used to develop a system model that can map the input function to the output. Supervised learning algorithms are extensively used to proffer solutions to problems pertaining to classification and regression [25]. Though supervised learning has the advantage of high convergence speed and action quality, large training data is required, and data labeling is performed manually, thus increasing the cost and complexity of data processing [26]. Examples of these algorithms include Decision trees, K-Nearest neighbours (KNN), Support vector machine, Linear regression, Gaussian process regression, and Support vector regression.
- **Unsupervised learning:** This learning method uses unlabeled input data to learn a function that can be used to explore the hidden pattern or features from datasets. Unsupervised learning techniques have the advantage of less or no human supervision compared to their supervised learning counterparts and are best known for solving clustering, outlier detection, and dimensionality reduction problems [27]. However, its main disadvantages lie in its low accuracy and higher computational complexity due to the unavailability of sample data-label pairs [28]. Examples of these algorithms include K-means clustering, Principal component analysis, Isometric mapping, Hierarchical clustering, and Independent component analysis.
- **Reinforcement learning (RL):** In contrast to supervised learning and unsupervised learning, RL is designed to enable multiple agents (e.g., IoT devices) to learn by interacting with their environment and are trained based on the reward (positive or negative) obtained

through a trial and error approach [29]. RL algorithms aim to learn optimal policy (i.e., optimal mapping of situations into actions) to maximize the cumulative reward over time [30]. This is achieved by striking an appropriate balance between exploration for new actions and exploitation of known actions [31]. Because RL algorithms can trade off immediate rewards for long-term gains, they are known to be suitable for decision-making, planning, and scheduling in stochastic network environments [32]. Nonetheless, for a large-scale network, the curse of dimensionality due to large state space may degrade the convergence rate of RL algorithms [33]. Examples of these algorithms include Multi-armed bandits, Markov decision process, dynamic programming, and Temporal-difference learning.

Another important ML technique that continues to attract significant attention in recent times is deep learning (DL). The traditional ML algorithms are limited in their applications for processing complex data-set representations, as they depend on adequate feature extraction to learn and perform various tasks [34]. Interestingly, DL algorithms, inspired by mimicking the biological nervous system (i.e., neural network), can perform feature extraction and transformation from unlabeled or unstructured data [35]. DL models comprise several multiple interconnected layers of neurons, with each layer using the output of the preceding layer as its input. This enables the DL model to provide a more refined estimate of a complex function [36]. Vast applications of DL algorithms have been recorded in different areas such as pattern recognition, natural language processing, computer vision, regression, etc [37]. Popular DL models extended to IoT networks include convolutional neural networks (CNN), recurrent neural networks (RNN), deep belief networks, and autoencoders. In addition, depending on how training data is used, DL can also be categorized as deep supervised, deep unsupervised learning, and deep reinforcement learning [38].

TABLE 2. List and definition of acronyms.

Acronym	Full Meaning
AE	Auto-encoder
AoI	Age of information
ANN	Artificial neural network
CNN	Convolutional neural network
DDPG	Deep deterministic policy gradient
DL	Deep learning
DNN	Deep neural network
DQN	Deep-Q network
DDQN	Double deep-Q network
DRL	Deep reinforcement learning
ELU	Exponential linear activation unit
ReLU	Rectified linear unit
EH	Energy harvesting
Exp3	exponential-weight algorithm for exploration and Exploitation
Exp4	Exponential-weight algorithm for Exploration and Exploitation using Expert advice
GAN	Generative adversarial network
ID	Information decoding
IoT	Internet of Things
LSTM	Long short-term memory
MAB	Multi-armed bandit
MDP	Markov decision processes
MEC	Mobile edge computing
ML	Machine learning
MIMO	Multiple-input-multiple-output
POMDP	Partially observable Markov decision process
PPO	Proximal policy optimization
RIS	Reconfigurable intelligent surfaces
RL	Reinforcement learning
RF	Radio frequency
RNN	Recurrent neural network
SD3	Softmax deep double deterministic policy Gradients
SARSA	State-action-reward-state-action
UAV	Unmanned aerial vehicle
UCB	Upper confidence bound
UE	User equipment
TD	Temporal difference
TD3	Twin-delayed deep deterministic policy gradient
TENG	Triboelectric nanogenerator

III. A COMPREHENSIVE REVIEW OF ML TECHNIQUES IN EH IoT NETWORKS

In this section, we discuss in detail various ML techniques employed in EH IoT networks following the ML classification method adopted in Section II. It is noteworthy that the application of classical supervised learning and unsupervised learning algorithms in EH IoT network optimization has received very little attention in the literature. This is attributed to the following factors: 1) due to the dynamic nature of EH IoT networks, labeling of training data becomes time-consuming and increasingly complex under supervised learning [39]; 2) aside from the higher computational complexity of unsupervised learning algorithms, their performance evaluation and validation can be challenging because of the lack of predefined output during training [40]; 3) Supervised and unsupervised learning algorithms require

large training data for improved performance; however, IoT devices have limited computational power, memory, and battery life to handle large volumes of data [41].

A few studies have attempted to investigate the performance of supervised learning algorithms [42], [43], [44], and unsupervised learning algorithms [45], in the context of EH IoT networks. In [42], the performance of linear regression and decision tree methods for predicting RF harvested energy is analyzed. Using the operating frequency and the harvested energy as the training data, a prediction accuracy of 80% is achieved under linear regression, while the decision tree yields an accuracy of 85%. Further on the decision tree, the study in [43], minimized the problem of transmission time in a cooperative communication system using uplink channel coefficients and harvested energy as the inputs to the proposed algorithm. In [44], the authors demonstrate the application of ensemble learning based on multiple decision trees to realize an accurate recognition pattern from a TENG-power sensor for security access purposes. An accuracy, of approximately 99% is realized during the training and the testing phase. In [45], an unsupervised Bayesian learning is proposed to learn the power consumption profile of an EH device in order to determine the amount of transmission power from the access point to be allocated for EH. Note, because of the aforementioned limitations of supervised learning and unsupervised learning, the solutions highlighted in the identified papers [42], [43], [45] are likely to perform poorly under a large-scale EH IoT network scenario. Hence, our main focus will be centered on ML algorithms that have been widely considered in this domain. These algorithms belong to the categories of RL and DL.

A. REINFORCEMENT LEARNING IN EH IoT NETWORKS

Having established that reinforcement learning enables an agent to maximize its cumulative reward by interacting with its environment, we now discuss in detail the different classes of reinforcement learning methods employed in EH IoT networks. These include the Multi-armed bandit problem, Markov decision process, Dynamic programming, and Temporal difference learning.

1) MULTI-ARMED BANDIT

Also referred to as the k -armed bandit technique, the multi-armed bandit (MAB) technique is conceived to solve a sequential decision-making problem given K possible actions. Each of the k actions selected by an agent (an IoT node) produces a reward defined by a fixed but unknown probability distribution [46]. The goal is to minimize the cumulative regret over a period of time by learning the reward performance of each action (exploration) and using the learned information to receive the best payoff (exploitation) [47]. This is a typical exploration-exploitation dilemma, where exploitation is deemed a reasonable choice to maximize the immediate expected reward, while exploration is considered a rational action that might yield a higher reward, in the long run [48].

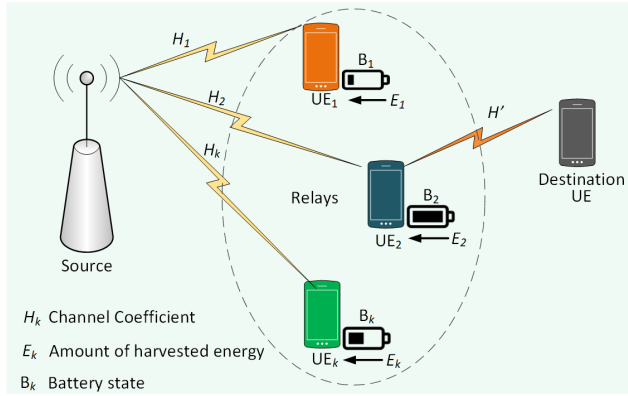


FIGURE 3. Cooperative EH IoT network.

To exemplify the MAB problem in the context of EH IoT, we illustrate a relay selection problem for throughput maximization using a single-agent MAB approach. As depicted in Fig. 3, we consider a single source serving multiple EH IoT nodes that can also be selected as a relay for a destination IoT device. The source is the agent, and the set of EH IoT relays $\mathcal{K} = \{1, \dots, k, \dots, K\}$ represents the set of arms. The transmission time is divided into slots $\mathcal{T} = \{1, \dots, t, \dots, T\}$. We denote the action selection of a relay by the source at slot t by a_t . We define the reward $r_{k,t}$ for selecting a relay as the achievable signal-to-interference plus noise ratio (SINR). Here, we assume that the source-relay link that yields the maximum SINR will maximize the throughput of the destination device. Hence, the goal of the source is to select the relay that will yield the maximum SINR following a policy $\mathcal{G} = \{a_1, \dots, a_t, \dots, a_T\}$ and eventually minimizing the possibility of data outages (i.e., regret) at the destination.

The regret function over the entire transmission time can be given as [49],

$$R(T) = \mathbb{E}_{\mathcal{G}^*} \left[\sum_{t=1}^T r_{a_t^*, t}^* \right] - \mathbb{E}_{\mathcal{G}} \left[\sum_{t=1}^T r_{a_t, t} \right], \quad (1)$$

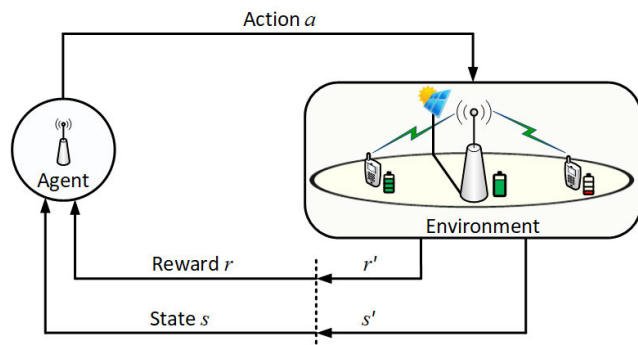
where $\mathcal{G}^*$ is the optimal policy, which knows the SINR of all relays and selects the optimal action a_t^* that yields the highest SINR. Following the expression in (1), the source aims to choose a relay across the time slots that will minimize the cumulative regret over a sequence of trials. This represents a typical exploitation-exploration problem because SINR of an EH relay depends on the amount of harvested energy $E_{k,t}$, battery state $B_{k,t}$, and channel condition $H_{k,t}$, which are unknown at the beginning of a transmission time slot, thus yielding a reward with different distributions. Different MAB learning algorithms have been developed to tackle this problem; among the popular ones are: (1) ϵ -greedy algorithm where an agent uses small probability ϵ to randomly explore different actions with equal probability each timestep and using $1 - \epsilon$ in each timestep to greedily select an action with the highest estimated average reward [50]; (2) Softmax algorithm where estimated action selection probabilities are used for choosing an action during

the exploration stage rather than randomly selecting an action as in ϵ -greedy case [51] (3) Upper confidence bound (UCB) algorithm, where the variance of the average estimate for selecting an action is adopted to simultaneously perform exploitation and exploration [52]; and (4) weighted policy method such as exponential-weight algorithm for exploration and exploitation (Exp3), and its modified version (Exp4) based on incorporation of expert advice [53].

Applications: MAB application in EH IoT networks has been demonstrated in [54], [55], [56], [57], [58], and [59] as summarized in Table 3. In [54], the authors develop an energy management system by varying the energy budget allocation of an EH sensor node for different tasks, which include data sampling, data reception, and data transmission. The goal is to maximize the collected information in a non-real-time manner. To address this, an Exp3 algorithm is proposed to determine the optimal combination of the energy budget for different tasks. Though the Exp3 outperforms the non-learning baseline techniques, its higher regret bound in a dynamic and complex environment remains its main limitation [60]. The Exp4 is designed to alleviate this challenge by incorporating expert advice in Exp3 algorithm. This idea is demonstrated in [55], to solve user association problem. In the study, the number of UEs, harvested energy by the small cell base station (SBS) and energy allocated to each user form the basis for the development of a set of experts that determines the best and available SBS. On selecting an SBS, the user reward is determined through the success transmission probability. The extension of this study is provided in [56] to deal with the associated complexity and convergence problem with many agents. To address this issue, the mean-field approximation technique combined with UCB is incorporated into the developed solution. The result obtained converges faster for a larger number of users. This is attributed to the capability of an agent in the mean-field game to ignore the spatial relationship between other agents. However, the combination of the two techniques may increase the computational complexity of the developed algorithm. Alternatively, adopting the idea of hierarchical arm selection with different quantization steps to improve the convergence rate of the UCB algorithm as demonstrated in [57] could be a viable solution. The study adopts the UCB to maximize the harvested energy and achievable SINR under channel uncertainties. In the study, the UCB algorithm enables the central controller to select a super-arm with the maximum upper bound, i.e., a combination of beamforming weights. The solution achieves maximum received power for EH, roughly at 80% of the ideal value with known statistical channel information. In another UCB-related work [58], the energy efficiency of the system is maximized by optimizing the downlink transmission power. In the study, due to the unavailability of channel state information (CSI), the source learns the optimal set of transmit power using the codeword transmitted in the uplink to determine whether the downlink transmission is successful or not. Unlike the aforementioned studies, the authors of [59], demonstrate the application of

TABLE 3. Summary of contributions to MAB applications in EH IoT networks.

Ref	Network scenario	EH Type	Main Objective	Input	Learning algorithm	Output	Main Findings	Remark/Limitation
[54]	Wireless sensor network	Solar	Energy budget optimization for maximizing sensor information collection	Sampled data, received data and transmitted data.	Exp3	Estimated information collection	Compared with the non-learning baseline methods with 98% performance, the long-term collected information based on the proposed solution improves by 120% in	Better performance can be achieved with Exp4 since it guarantees a lower regret.
[55]	Small cell network	Not specified	Enhancing UE association strategy for seamless user connectivity	Number of UEs, harvested energy by SBS, and allocated power.	Exp4	Near-optimal user association	The success probability of UE association approaches the exhaustive search method with known statistical information	Incorporating the load balancing constraints is required in the design to mitigate overloading of SBS.
[56]	Small cell network	Not specified	User association strategy for enhanced user connectivity	Number of UEs, harvested energy by SBS, and allocated power	UCB	Near-optimal user association	Solution converges faster with higher numbers of users compared with greedy-based algorithms	Analysis of trade-off between the convergence rate and computation complexity is not captured.
[57]	MIMO network	RF	Beamforming weight optimization for EH maximization	Beamforming weights	UCB	Maximized harvested energy	The maximum power received for EH is estimated at 80% of the ideal value with known CSI.	The algorithm is developed to operate as a centralized coordinator. This may not support network scalability.
[58]	Wireless powered network	RF	Transmit power optimization for energy efficiency maximization	Harvested energy, weighted sum-rate and a set of power allocation.	UCB	Estimated energy efficiency	The scheme with known CSI only outperforms the proposed algorithm at -60 dBm transmit powers which has negligible effect in practice	The solution is limited to a simple scenario, which may not be effective under a complex scenario, e.g., multiple sources with multi-antenna.
[59]	Underwater sensor network	Tides	Optimal contention window size for throughput maximization	Minimum and maximum contention window size values	ϵ -greedy	Estimated throughput	The realized throughput approaches the Branch and Bound method, where global network information is employed.	Unlike the UCB, which can perform exploration and exploitation simultaneously, the ϵ -greedy method cannot.

**FIGURE 4.** Markov decision process.

ϵ -greedy algorithm for solving the throughput maximization problem in a random access-based internet of underwater things (IoUT) network. The main objective is to enable the IoUT to select an optimal contention window size based on network data to be transmitted and harvested energy. As ϵ tends to zero, the realized throughput approaches that of the Branch and Bound optimization method where global network information is employed.

2) MARKOV DECISION PROCESSES

The practical RL problems are frequently formulated as a function of state-dependent quantities [61], unlike the MAB

problems, where stateless approaches can be employed to simplify designs using only the action-reward pairs [62]. The Markov decision process (MDP) offers a mathematical framework for sequential decision-making problems under uncertainty in RL [63]. An MDP can be represented by a tuple (S, A, P, R) where S and A denote finite sets of an agent's states and actions, respectively, P is the state transition probability from state $s \in S$ to a new state $s' \in S$ after taking an action a , and $r \in R$ is the immediate reward obtained by an agent from an environment after executing an action a . An illustration of this process is depicted in Fig. 4. At each sequence of discrete time steps $t = 0, 1, 2, 3 \dots$, the agent, which in this scenario is the access point, is in a particular state. The state elements can be the harvested energy from solar, its battery level, and the channel coefficients between the access point and the UEs. Based on these states, the agent selects any available action, which could be the chosen level of power allocation to the UEs. This process can be represented by a state-action pair (s_t, a_t) . At the next time epoch $t+1$, the system is transitioned into a new state s_{t+1} following a policy π while receiving a reward of r_{t+1} for choosing a_t from s_t . In the illustrated scenario, the reward may be modeled to minimize the energy consumption of the access point while the agent continues to improve its power allocation policy to maximize the reward.

A policy π , which may be a deterministic policy or a stochastic policy, determines how an agent chooses an action in different states. In the former case, the policy is denoted by $\pi(s) := a$, indicating the action that must be selected in a state, while in the latter case, denoted by $\pi(s, a) := P(a|s)$, represents a mapping from a state to the probability of choosing an action. Thus, the corresponding value function of a state s under a policy π which quantifies cumulative reward when starting from s_0 following a policy π can be expressed as:

$$v_\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1}) \mid s_0 = s \right], \quad (2)$$

where $\gamma \in [0, 1]$ is the discount rate. Similarly, the measure of taking action a in state s then following a policy π onward (i.e., the action-value function) can be given as:

$$q_\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1}) \mid s_0 = s, a_0 = a \right]. \quad (3)$$

Therefore, the main goal of the agent is to find the optimal policy π_* that maximizes $v(s)$ i.e., the optimal state-value function denoted as $v_*(s)$. This can be derived recursively without referencing any particular policy following Bellman's optimality equation, which can be written as:

$$v_*(s) = \max_a \sum_{s', r} p(s', r|s, a) [r + \gamma v_*(s')]. \quad (4)$$

For the optimal action-value q_* its associated Bellman optimality equation can be given as:

$$q_*(s, a) = \sum_{s', r} p(s', r|s, a) [r + \gamma \max_{a'} q_*(s', a')]. \quad (5)$$

Note that in the MDP model, it is assumed that the agent has the perfect knowledge of the state of the system. However, this assumption may become invalid in scenarios where certain features of the state are hidden from the agent [64]. Interestingly, the state uncertainty constraints can be relaxed using an extension of the MDP framework known as the partially observable Markov decision process (POMDP). A POMDP differs from MDP in that the agent perceives an observation $o \in \mathcal{O}$, that may uncover some information about s' instead of observing s' directly. Hence, the agent maintains the belief state $b \in \mathcal{B}$ as an estimate of its state [65], [66].

Since the main task of RL is to obtain an optimal policy that maximizes the long-term reward, several RL algorithms based on MDP/POMDP value functions have been developed to achieve this purpose. These algorithms can be broadly categorized under model-based RL and model-free RL. In the former approach, the agent learns the transition dynamics $p(s'|s, a)$ referred to as models of the environments to improve its policy [67]. These algorithms have a higher learning rate than model-free algorithms because they can reuse the stored data from their simulated models. Nonetheless, because of their substantial computational overhead and reliance on the assumption of a perfect model, their applications are

less employed [68]. In the model-free approach, the agent learns and refines its policy by interacting directly with the environment without prior knowledge of the model of the environment. Examples of model-based RL algorithms belong to the class of dynamic programming, e.g., policy iteration, value iteration and value function approximation. For temporal-difference learning, algorithms under this classification can be regarded as model-free algorithms, e.g., Monte Carlo, Q -learning and SARSA. In the following, we present the applications of these algorithms in the context of EH IoT networks.

3) DYNAMIC PROGRAMMING AND ITS APPLICATIONS

Though dynamic programming algorithms are rarely used in the EH IoT network due to the limitations highlighted above, high preference has been given to value iteration and value function approximation over policy iteration, especially when the state space is large. This is due to the advantages of their fast convergence and ease of implementation [69]. To demonstrate dynamic programming application, the authors of [70] develop a value iteration-based control policy for an EH secondary UE in a cognitive radio transmission to optimally perform channel sensing, probing, and data transmission. The goal is to maximize the transmission data rate. Moreover, the concept of the after-state, also known as post-decision state (PDS) is incorporated in the design to avoid action exploration while minimizing the dimension of state-action space. To further reduce the state space, the time slot partitioning is decomposed into two for channel sensing and probing and data transmission PDS updates. In [71], the authors highlight that the after-state-based learning approach can be further accelerated by performing multiple PDS updates in a single time slot. To achieve this, the concept of virtual experience is introduced, where the statistical data extrapolated from the data arrival and energy packet arrival are used to update the PDS with different data buffers and battery states. Further, to reduce the computational complexity introduced by the virtual experience, the PDS value functions are approximated using the value function approximation approach. A summary of the aforementioned studies is provided in Table 4.

4) TEMPORAL DIFFERENCE (TD) LEARNING

In the TD learning approach, the model of the environment is not required; however, the agent learns time-delayed rewards by interacting directly with the environment [72]. Unlike the Monte Carlo method which is restricted to episodic tasks, the TD method continuously learns a good estimate of expected reward by bootstrapping from the current estimates of the expected return [73]. TD algorithms can be categorized based on the policy control method, i.e., on-policy or off-policy [74]. State-action-reward-state-action (SARSA) and Q -learning are the popular on-policy-based and off-policy-based algorithms, respectively. The SARSA algorithm updates its Q value using its actual policy action a' as given

TABLE 4. Summary of dynamic programming solutions to MDP problems in EH IoT networks.

Ref	Network scenario	EH Type	Main Objective	Input	Learning algorithm	Output	Main Findings	Remark/Limitation
[70]	Cognitive radio network	Solar, wind, and thermal	Channel sensing, estimation, and data transmission control.	Time slots allocated sensing, probing, and transmitting	Value iteration	Optimized allocated time slot	Learning rate and average data transmission outperform the solutions where average channel gain and purely greedy policies are considered.	It is assumed that the perfect state transition probability is known by the secondary user, which may not be realistic.
[71]	wireless sensor network	Not specified	Optimal packet transmission scheduling policy for queuing delay minimization	Battery capacity, harvested energy and channel coefficient	Value function approximation	Smaller transmission delay rate	Faster learning rate is achieved compared to the baseline Q-learning and SARSA	The solution is based on the use of extrapolated data from energy and data arrival, which may be unreliable due to their causal nature

in (6) [75],

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma Q(s', a') - Q(s, a) \right], \quad (6)$$

where α is the learning rate. Q -learning algorithm updates its Q -value based on the action that maximizes the estimate of Q in the next epoch, as given in (7) [76],

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_a Q(s', a) - Q(s, a) \right]. \quad (7)$$

In general, Q -learning finds an optimal solution more quickly than SARSA, albeit with poor online performance. Nevertheless, because SARSA is maximizing on-policy rewards, it provides safer actions for the agent [61].

- *SARSA and its applications:* Application of SARSA in EH IoT networks has been demonstrated in [77], [78], [79], [80], and [81] as summarized in Table 5. In [77], the authors develop a power consumption management algorithm for solar-based EH sensors. The proposed algorithm enables the EH sensor to select the appropriate duty cycle based on the state of battery level, weather, harvested energy, and the deviation from energy neutrality. The goal is to minimize the deviation from the sensor's battery level at the end of an episode with respect to a reference battery level at the beginning of the episode. A near-optimal solution with a 6% deviation from energy neutrality is achieved. However, the study considers a finite number of states, which may be impractical as the number of state elements can be infinite. For instance, the study in [78], captured the channel state and data buffer level in addition to some of the state elements considered in [77]. Thus, to deal with the possibility of infinite states, the authors introduced the linear function approximation method, which uses the sum of the weighted features obtained from the mapping of the state-action pairs to a feature value to update the Q -values. The goal of the study is to develop an optimal transmit power policy that maximizes the throughput at the destination receiver. A 6% deviation in throughput is observed in comparison with the upper bound off policy. The linear function approximation approach has also been

extended to studies presented in [79] and [80]. In [79], the authors develop and compare the performance of centralized and distributed algorithms in a multi-user scenario. In the former, the state information of all nodes is processed at a central unit, whereas the state information of neighboring nodes is employed in the latter. The goal is to solve the problem of transmission delay through transmission power control and routing selection policies. A negligible difference in the convergence rate of the proposed centralized and distributed algorithms is observed. In [80], the authors derived a power allocation policy that maximizes the throughput in a MIMO system. Unlike [77], [78], and [79] which are based on ϵ -greedy for action selection, an adaptive softmax-greedy scheme is developed in the study to improve the power allocation policy and convergence rate. An obvious similarity in [77], [78], [79], and [80] is that the problems are formulated using the MDP framework, where full certainty of the state is assumed. However, this assumption cannot hold in a partially observable environment. To overcome this constraint, the application of POMDP framework is demonstrated in [81] where EH sources and relay nodes only have partial knowledge of harvested energy, channel gains, data buffer levels, and battery levels. The goal is to develop a power allocation policy that maximizes the throughput at the destination node. To alleviate the complexity of POMDP, Kalman's filter-based channel predictor is incorporated in the design for channel coefficient estimation, while the remaining state information is obtained via information exchange between the cooperative EH nodes.

- *Q-learning and its applications:* Application of Q -learning in EH IoT networks has been demonstrated in [83], [84], [85], [86], [87], and [88] as summarized in Table 6. In [83], the authors develop a duty cycle policy that serves as an input to a sensor's on-off scheduler to balance EH and energy consumption rate. A faster convergence rate is achieved compared with the closed-form solution based on the continuous-time Markov chain. In [84], a reliability function to improve the prediction accuracy of solar-based EH system is

TABLE 5. Summary of SARSA applications in EH IoT networks.

Ref	Network scenario	EH Type	Main Objective	Input	Output	Main Findings	Remark/Limitation
[77]	Wireless sensor network	Solar	Duty cycle control for minimizing deviation from energy neutrality.	battery level, harvested energy, and the deviation from energy neutrality	Near-optimal Duty cycle estimation	A near-optimal solution with 6% deviation from energy neutrality is achieved compared with off-policy baseline methods with over 23% deviation.	The study employs historical solar data for pre-training purposes; however, this may increase the computational overhead.
[78]	Cooperative communication network	Not specified	Power allocation policy optimization for throughput maximization	Channel coefficient, harvested energy, battery level, and data buffer size	Power allocation estimation	A 6% reduction in throughput is observed in comparison with the optimal off policy.	The study adopts a linear function approximation for estimating Q -values which may introduce an irremovable bias as the optimal Q function may be non-linear [82]
[79]	Wireless sensor network	Solar	Optimal transmission power control and route selection policy for transmission delay minimization	Battery level and capacity, harvested energy, sensed data and channel coefficient	Estimated transmission power	Negligible difference in the convergence rate of the proposed centralized and distributed algorithm is observed.	Incorporation of the principle of after-state in the distributed and centralized models can increase the convergence rate of the solution.
[80]	MIMO network	Wind and solar	Optimal power allocation policy for throughput maximization	Channel coefficient, battery level and harvested energy	Estimated power allocation	Achieves faster convergence compared with marginal deviation in throughput from the upper bound methods	The improved convergence rate due to the use of hybrid ϵ -greedy and softmax increases computational complexity
[81]	Cooperative communication network	Solar	Optimal transmission power policy for throughput maximization.	Harvested energy, channel gains, data buffer levels, and battery levels.	Estimated transmission power	Higher throughput is achieved compared with cases where hasty policy, central SARSA, and non-cooperative learning are employed	The model is prone to high signaling overhead due to reliance on information exchange between local nodes to overcome POMDP complexity

developed. Using a daily profile approach, the predicted harvested energy in different time slots is evaluated as a function of the prediction error, estimated reliability level, and a weighting factor to give higher emphasis on the current time slot. Moreover, a modified learning rate is formulated based on the difference between overall and current prediction error ratios, such that the Q value is significantly reduced when the prediction error rate is high. However, a common constraint of [83] and [84] is the impact of fading transmission channels that is unaccounted for. This limitation is captured in an RF-based EH system [85], where a joint policy for information transmission rate and EH interval is optimized with the goal of maximizing the throughput. Though the proposed algorithm outperforms the model-based methods in terms of achievable throughput and convergence rate, the MDP frame adopted is oversimplified as the channel gains are the only state element considered. This limitation is addressed in a related study [86], where the uplink and downlink channel gains, and battery level of the receiver are captured in the state space. The main focus is to develop a channel assignment policy for information decoding (ID) and EH processes to minimize the outage probability in the network. At a lower convergence time, the proposed solution approaches the solution based on exhaustive search algorithms. Unlike [83], [84], [85],

and [86] that are based on small state-action pairs, and a finite state space, the authors of [87], examine a mobile edge computing (MEC) network with a large dimension of state-action and a continuous state space. The goal is to develop a computational offloading policy that enables an EH device to decide whether to initiate data processing, execute local computation, or offload its data to the MEC server. The after-state approach is employed to minimize the dimension of the state-action pair, while the value function approximation method is adopted to realize a finite state space. In another related paper [88], an optimal MEC server request policy for the age of information (AoI) minimization is demonstrated. In the study, the server responds to the destination request either by demanding a fresh measurement from the EH sensors or using the aged data in its cache. The goal is to balance the tradeoff between energy consumption and data staleness costs. Unlike the aforementioned papers, the study further examines the scenario where only partial knowledge of the battery level is available in the MDP model. A 30% deviation from the value-iteration and Q-learning-based solutions with exact knowledge of battery level is observed. It is noteworthy that the standard Q -learning is prone to overestimation because the max operator uses the same value to both choose and evaluate an action. To combat this problem, the double Q learning

(DQL) is developed [89]. In the context of EH IoT networks, this idea is demonstrated in [90]. In the study, an energy management controller employs the DQL model to realize an optimal duty cycle for sleep and wake-up time operations. A 98.5 % success rate is achieved based on the evaluation of the completed and missed transmissions in the system. Moreover, the results outperform static period and fuzzy logic-based controllers.

B. DEEP LEARNING

The notion of deep learning originates from the artificial neural network (ANN) which consists of multiple groups of neurons at each layer of processing units. As depicted in Fig. 5(a) a typical ANN, also referred to as a shallow neural network, uses zero to three hidden layers between the input and output layers [91]. The computation at each layer of ANN involves a weighted sum of input values in each neuron, which is constrained by a non-linear activation function that enables a neuron to generate an output when the inputs satisfy a certain threshold. This forward propagation process is depicted in Fig. 5(b), and the mathematical representation is given in (8):

$$y_j = a\left(\sum_{i=1}^m w_{ij}x_i + b_j\right), \quad (8)$$

where x_j , y_{ij} , w_{ij} , and b_j denote the input activation, output activation, weight, and bias, respectively, and $a(\cdot)$ is the activation function. ANNs are trained to minimize the mean square error between output and the actual output of the network. This can be achieved by calculating the gradient of the loss function, which is used to adjust the weights, a process referred to as backpropagation [93]. The forward and backward propagation cycle is performed iteratively until the network is able to give an accurate estimate of the output in most trials. The application of ANN is illustrated in [94] for the feature extraction process on the triboelectric signals generated from a TENG-powered insole sensor. The goal is to use the extracted features for waist motion and gait analyses to aid patient classification and to develop a waist and lower-limb rehabilitation plan. A 98.4% patient recognition accuracy is achieved.

ANNs have relatively simple architecture; however, they are limited in their use for modeling and representation of complex and hierarchical features. This triggers the need for deep architectures comprising tens or hundreds of hidden layers (generally known as deep artificial neural networks (deep ANNs)), to combat the limitations of ANN. An illustration of Deep ANN is shown in Fig. 5(c). The illustration depicts a deep ANN network fed with channel coefficient $\{h_1, h_2, \dots, h_L\}$ from the BS to the UE $\{UE_1, UE_2, \dots, UE_N\}$, the harvested RF energy $\{e_1, e_2, \dots, e_N\}$ by the UEs, and the battery levels $\{b_1, b_2, \dots, b_N\}$ of the UEs, as inputs. Based on the magnitude of the input parameters, and after repeated process of

neuron activation and backpropagation, the network provides an estimate of the corresponding outputs, which in this case represent the power allocation factor $\{\alpha_1, \alpha_2, \dots, \alpha_N\}$ and power splitting ratio $\{\rho_1, \rho_2, \dots, \rho_N\}$. Depending on the use case, deep architectures can be grouped under discriminative, generative, and hybrid deep architectures [95]. Discriminative deep architectures rely on supervised learning approach to provide a discriminative function in classification and regression applications. Examples of DNNs in this category are deep ANN and CNN. The generative deep architectures aim to provide high-level correlation properties of the observed data for pattern analysis or synthesis purposes. Contrary to the discriminative architectures, in generative deep architectures, data labeling is unimportant; hence they can be employed in unsupervised pretraining for supervised learning tasks. Examples include RNN, auto-encoders, generative adversarial networks (GAN), Restricted Boltzmann machines (RBM) etc. The hybrid deep architectures are composed of generative and discriminative deep architectures, especially to perform discriminative tasks, but enhanced by the optimized outcomes of the generative architecture. The reader can refer to [96], for more details on various sub-classes of the aforementioned deep learning architectures. Applications of DL has also been extended to RL. Details on this is presented in Section III-C. In addition, it is noteworthy that DNNs may become difficult to train due to their saturated output caused by the vanishing gradient problem. To combat this problem, various activation functions have been developed, such as the ones in the categories of rectified and exponential activation functions [97]. Also, another fundamental issue that is worth mentioning is the problem of overfitting. This is associated with a high variance in test error as a result of a change in the training dataset [98]. Various regularization methods have been proposed to address this problem. These include L^p regularization, dropout, and early stopping. In the following, we shall focus on the popular deep learning models adopted in EH IoT networks.

1) DISCRIMINATIVE (SUPERVISED) DL

In the following, we discuss the notable discriminative DL techniques in EH IoT networks.

- *Deep ANN and its applications:* Application of deep ANN in EH IoT networks has been demonstrated in [99], [100], [101], [102], and [103] as summarized in Table 7. In the context of regression-based problems, the authors of [99] develop a deep ANN that optimizes the transmit power and power splitting ratios in a wireless-powered D2D network. Using the channel gains as the input parameters to the neural network, the goal is to maximize the system sum-rate while minimizing the energy outage probability. Further, the input channel vector and the EH-constrained cost function are modified to avoid the training failure caused by the non-representation of channel magnitudes and the possibility of zero gradient computation, respectively. From the viewpoint of different multi-variable inputs,

TABLE 6. Summary of Q-learning applications in EH IoT networks.

Ref	Network scenario	EH Type	Main Objective	Input	Output	Main Findings	Remark/Limitation
[83]	Wireless sensor network	Solar	Optimal duty cycle policy for achieving energy neutrality	Buffer capacity, sleep/awake state, and link quality	Estimated duty cycle	A faster convergence rate for an optimal adaptive rate duty cycle is observed compared to the closed-form method	An oversimplified MDP framework is adopted. For instance, the impact of fading channels is not captured in the state elements.
[84]	wireless sensor network	Solar	Development of reliability level for EH prediction.	Harvested energy profile from historical data	Near-optimal EH prediction	The proposed solution outperformed the baseline method where the present weather conditions are not considered.	The non-consideration of discounted reward in the proposed Q-function suggests an infinite reward system, which may not be realistic.
[85]	wireless powered network	RF	Transmission rate and energy harvesting interval control for throughput maximization	Channel gain and harvested energy level	Optimized EH rate and transmit time allocation	A lower complexity and faster convergence are observed in the proposed solution compared with the model-based methods.	Only the channel coefficient is considered as the state parameter whereas, other elements, such as battery level, can be incorporated in the state function.
[86]	Wireless powered network	RF	Channel allocation policy for outage probability minimization	Battery level, uplink and downlink channel gains	Accurate channel estimation	Results approaches the solution under the exhaustive search algorithm with fast convergence rate	The solution is based on a finite state element, which may not be applicable where the state space is infinite
[87]	MEC-based network	Not specified	EH and generated data prediction for developing optimal offloading policy	Data and energy buffer levels, channel gains, and computation latency	Estimated EH and data generation	Lower number of dropped data and transmission latency is observed compared with Q-learning and Myopic methods.	Performing multiple after-state updates in a single time slot can further accelerate the learning rate of the developed algorithm.
[88]	MEC-based network	Not specified	Optimal MEC server data request policy for average AoI minimization	Battery level and AoI in the local cache	Accurate AoI estimation	Q-learning with partial knowledge of battery state shows 30% deviation from the RL methods with full knowledge of battery level.	The study can be enhanced by considering the partial knowledge of energy arrival rate and channel gains.
[90]	Wireless sensor network	TEG	Optimal duty cycle policy for energy consumption management	Sleep time and battery level	Estimated sleep and wake-up time	Results outperform static period and fuzzy logic-based controllers while achieving a 98.5% transmission success rate.	Incorporating additional practical elements in the state space, such as channel gains, will enhance the practicality of the study.

the authors of [100], develop a neural network that evaluates the system throughput based on the number of selected relays, position and number of power beacon antennas, residual noise, etc. Moreover, the study adopts the exponential linear activation unit (ELU) to tackle the diminishing gradient problem associated with the rectified linear unit (ReLU), as proposed in [99]. In another related paper [101], the outage probability in an EH UAV-assisted cognitive NOMA network is analyzed for several parameters such as UAV height, radius, angular motion, target rates of the UEs, etc. Moreover, the study introduces Adaptive Moment Estimation (ADAM), a variant of the gradient descent method, to minimize the computational complexity that scales with training data size. However, the power splitting factor of the UAV, which is critical to the outage probability evaluation, is not considered in the inputs to the DNN. To minimize the training overhead, the authors of [102] propose an offline training strategy wherein

the sampled output optimization variable from the sequential parametric convex approximation technique is used for the pretraining process. The study aims to jointly optimize the time allocation for EH and ID, power allocation, and transmit beamforming vector to minimize the transmit power in the downlink. Further, the study introduces the application of the dropout technique in tackling the problem of overfitting. However, since a large amount of training data is required to improve the system accuracy, data labeling may become expensive. This challenge is addressed in [103], where the authors employ training data obtained from the Lagrangian iterative solution to perform a supervised pretraining. The goal is to initialize the weights of the DNN needed to estimate the transmit power and power splitting ratios. Further, an unsupervised training approach is used to adjust the weights to achieve optimal values, which are later used to maximize the system sum-rate.

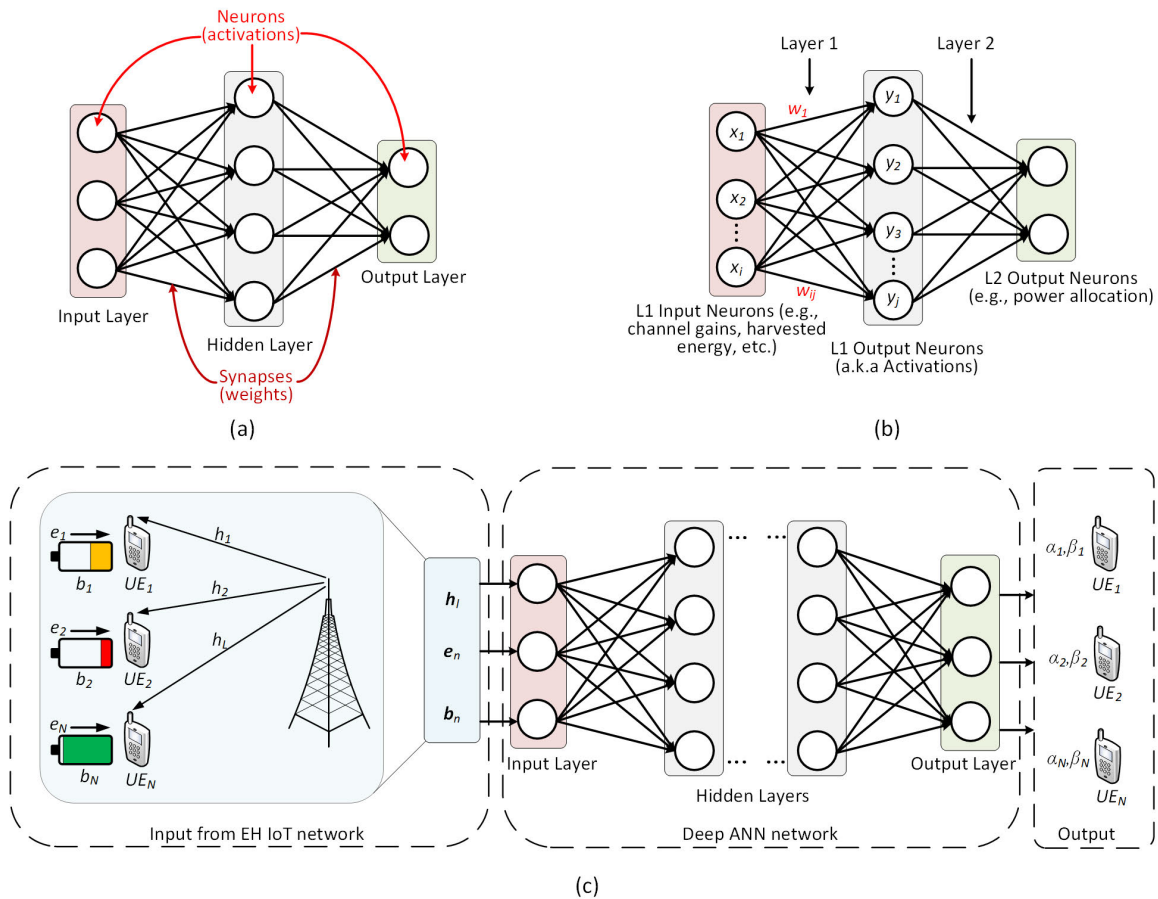


FIGURE 5. Neural networks (adapted from [92]). (a) ANN architecture. (b) ANN data propagation. (c) Deep ANN architecture for RF-based EH IoT Network.

- **CNN and its applications:** The deep ANN suffers from the various problems associated with high dimensionality due to the full connectivity of neurons from one layer to the next layer. Moreover, it does not consider the spatial representation of data. Thus, it exhibits poor performance when larger and more complex input data, such as an image signal, needs to be processed. Therefore, CNN, which provides sparse connectivity, equivariant representations, and parameter sharing [104], was developed to overcome these limitations. CNN architecture, as depicted in Fig. 6, comprises three distinct types of layers, i.e., convolution, pooling (subsampling), and fully connected layers. The convolution layer applies kernels (filters), which are convolved with the input tensor to generate a feature map. The convolved output is then passed through a nonlinear activation function. The expression to calculate the value of a feature element, h_{ij} , in a k -th feature map can be written as [106]:

$$h_{i,j,k}^l = a(\mathbf{w}_k^l \mathbf{x}_{i,j}^l + b_k^l), \quad (9)$$

where $\mathbf{w}_k^l$, $\mathbf{x}_{i,j}^l$, and b_k^l denote the weight vector, input patch, and bias term, respectively, and $a(\cdot)$,

is the activation function. The pooling layers are usually placed after a convolution layer to reduce the resolution of the feature map using the average or max pooling strategy [107]. Hence, this operation leads to a decrease in the parameters in the network, reduces computation time, provides translation invariance to small shifts and distortion, and thus, controls overfitting. After several convolution and pooling operations, the final output feature maps are transformed (flattened) into a one-dimensional (1D) array, and connected to one or more fully connected layers, as seen in the conventional ANN. The last layer of CNN is the output layer. This layer performs classification tasks in which a probability value is assigned to each class. The softmax and SVM operators are commonly used for this purpose.

Applications: Notable applications of CNN have been demonstrated in [105], [108], [109], [110], as summarized in Table 8. CNN is known to be suitable for image classification tasks, interestingly, the channel estimation problems in the EH IoT network can also be transformed into image-processing problems. This concept is demonstrated in [105] to improve power allocation for EH in reconfigurable intelligent surfaces (RIS)-based

TABLE 7. Summary of deep ANN applications in EH IoT networks.

Ref	Network scenario	EH Type	Main Objective	Input	AF	Output	Main Findings	Remark/Limitation
[99]	D2D network	RF	Optimization of transmit power and power splitting ratios for sum-rate maximization and EH outage minimization	Channel gains	ReLU	Estimated channels coefficients	At a lower computation time, results approach the Lagrangian-based iterative solution and exhaustive search method.	The resource allocation design is solely based on channel gains. Incorporation of different network variables as input can enhance the solution.
[100]	Cooperative NOMA network	RF	Throughput prediction	Selected relays, position and number of beacon antennas, etc.	ELU	Estimated throughput	The predicted results from deep ANN compare favourably the ones obtained from simulations and closed-form analyses.	A relay selection based on the mean channel gains of the source-relay-destination link could provide a better throughput performance.
[101]	UAV-aided cognitive network	RF	Outage probability prediction	UAV height, radius, angular motion, target rates of the UEs etc.	ELU	Near optimal outage probability prediction	Achieves lower computational time and is in line with results from a simulation and analytical solutions.	Power splitting factor, a key input parameter to the DNN for data outage prediction, is not considered.
[102]	Wireless powered network	RF	Joint optimization of time and power allocation, and transmit beamforming vectors	Channel gains	ReLU	Estimated power and time allocation, and beam-forming vector	A lower downlink transmit power is achieved compared with the results obtained under the traditional iterative algorithm.	The pretraining process, incorporated in the design may increase the trade-off between convergence rate and computational complexity.
[103]	Wireless power network	RF	Transmit power and power splitting ratio optimization for sum-rate maximization.	Channel gains	ReLU	Near-optimal power allocation and power splitting ratio estimation	A faster convergence rate is observed compared with the cases where Lagrangian-based iterative and DNN methods are solely used.	Higher computational complexity remains the main challenge due to the adopted pretraining process.

AF - Activation function.

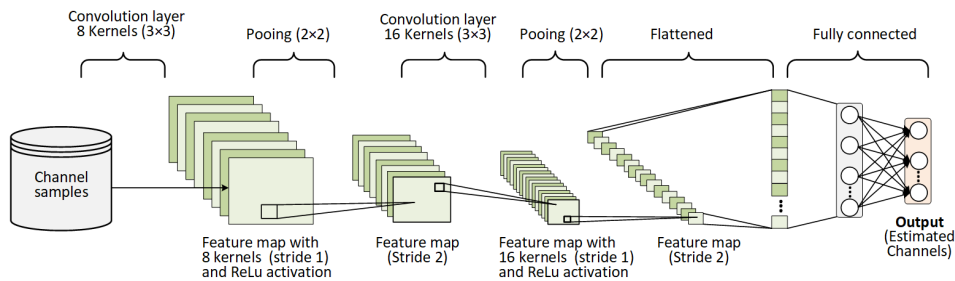


FIGURE 6. CNN architecture for channel estimation in EH network (adapted from [105]).

UAV-aid cell-free MIMO system. The results obtained outperformed the traditional least squares and least minimum mean square error approaches. However, the optimization of power allocation to maximize the harvested energy based on the estimated channels is not addressed. In [108], the authors employ training data from derived genetic-based algorithms to predict and minimize outage probability. To achieve this, optimization of allocated time slots for EH and ID coupled secrecy rate constraints are incorporated in the outage probability minimization problem. The CNN solution achieves a lower convergence time with a negligible mean squared prediction error compared to

the genetic-based algorithm. Due to their similarity with external stimuli, triboelectric signals have also been exploited for various pattern recognition functions. In [109], time and frequency domain signals transmitted from a TENG-powered sensor are processed for monitoring tasks. The goal is to exploit the signals for bridge deformation identification and rockfall detection. After 50 iterative training steps, a 100% recognition accuracy is achieved. In another related work [110], the authors demonstrate the application of ADAM optimizer to improve the training convergence rate in a hand gesture recognition task. A 98.57% recognition accuracy is achieved.

TABLE 8. Summary of CNN applications in EH IoT networks.

Ref	Network scenario	EH Type	Main Objective	Input	Output	Main Findings	Remark/Limitation
[105]	RIS-UAV aided networks	RF	Channel estimation for power allocation processes.	Channel coefficients from the UAV and RIS	Estimated channels	Achieves better performance in comparison with least squares and least minimum mean square error approach.	Power allocation optimization is not considered.
[108]	UAV-aided networks	RF	Data and secrecy outage probability minimization.	channel coefficients, TS factor and user coordinate.	Estimated channels and optimized TS factor	Achieves a lower convergence time with a negligible mean squared prediction error compared to the genetic-based algorithm.	Employing PS or hybrid PS-TS can provide additional improvement due to the latency issue in TS-based systems.
[109]	Wireless sensor network	TENG	Bridge deformation and rockfall detection.	Time and frequency-based triboelectric signals	Refined signal for various categories of deformation and rock mass	Achieves 100% accuracy after 50 iterative training steps.	Details on the method of neuron weight refinement e.g., backpropagation, are not included.
[110]	Wireless sensor network	TENG	Pattern recognition for hand gestures.	Time series triboelectric signals	Refined signals for different hand gesture recognition pattern	Achieves 98.57% hand gesture recognition pattern.	Considering a frequency domain-based signal could minimize the complexity of the recognition model.

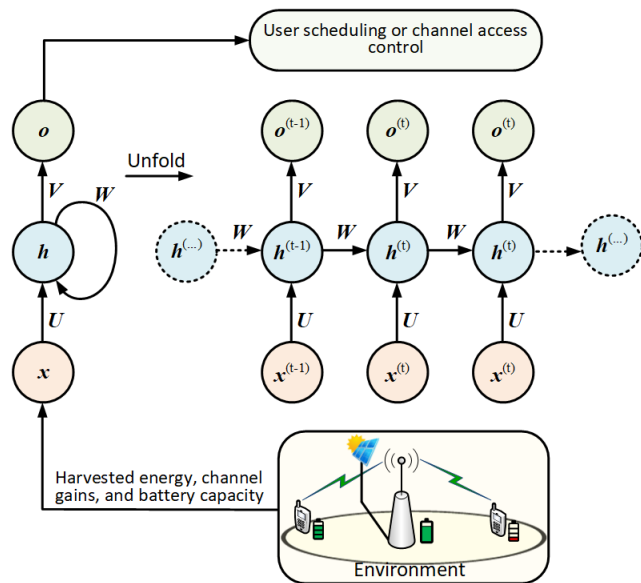


FIGURE 7. RNN structure (adapted from [104]).

- *RNN and its applications:* Unlike the deep ANN and CNN, which can be regarded as feedforward neural networks, the RNN consists of a cycle of processing units. It uses time series or sequential data in which the output o of the previous layer becomes the input x of the current layer, which is the only layer in the network but represented in different time series. Consequently, this results in a feedback loop where the output of one layer is supplied as input to itself. This enables the network to possess memory of previous inputs, which allows it to influence current input and output. An illustration of the standard RNN is depicted in Fig. 7. The network may be designed to perform user scheduling or channel access control based on the network inputs such as

harvested energy by the access point, channel gains and battery level of the network nodes. Mathematically, the expression for the standard recurrent cell and its corresponding output can be given in (10) and (11), respectively [104].

$$h^{(t)} = \tanh(W h^{(t-1)} + U x^{(t)} + b), \quad (10)$$

$$o^{(t)} = c + V h^{(t)}. \quad (11)$$

The parameters U , V , and W denote the weight matrices for input-to-hidden, hidden-to-output, and hidden-to-hidden connections, respectively. The bias vectors are denoted by b and c , and $\tanh(\cdot)$, is the activation function. However, a major problem of RNN is the issue of vanishing and exploding gradients, which makes learning long-term dependencies in data sequence challenging [111]. To combat these problems, a Long Short-Term Memory (LSTM) cell is developed and introduced into the recurrent connections [112]. The LSTM cells store the information over time and regulate the flow of information by gated units [113]. To the best of the authors' knowledge, there are no known studies on the sole application of RNNs in EH IoT systems. However, they have been combined with RL algorithms to solve different problems in this domain. These investigations are captured in Section III-C, where deep RL techniques are discussed.

2) GENERATIVE (UNSUPERVISED) DL

In the following, we discuss the notable generative DL techniques in EH IoT networks.

- *Auto-encoder (AE):* As depicted in Fig. 8, an AE is a type of neural network model that consists of two parts, namely an encoder and a decoder. The encoder, formed by the input and the hidden layer is fed with

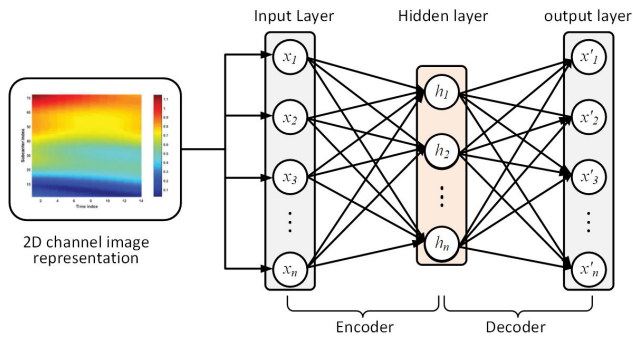


FIGURE 8. Conceptualized AE architecture of an RF-based EH IoT system. The 2D channel image representation [114], serves as input x to the input layer, h is the hidden layer, and x' is the reconstructed output.

an input signal such as 2D transmission channel image representations of an EH IoT network and maps the input into lower-dimensional encoding space. The decoder, formed by the hidden layer and output layer, transforms the signal back to the original space and reconstructs the original representation as accurately as possible [115]. This is achieved by minimizing the reconstruction loss. Since the hidden layer performs the representation learning, to ensure that learning is achieved, different types of constraints are imposed on the properties (such as size, activation, distribution, etc.,) of the hidden layer [116], [117]. Examples of AEs developed to implement these constraints include denoising AE, sparse AE, variational AE and so on [118]. A notable application of AE in the context of EH IoT network design is demonstrated in [119]. In the study, the authors employ variational AE, which leverages the node and energy distribution to dynamically determine and identify nodes with sufficient harvested energy that could serve as a potential relay.

- **Generative adversarial networks (GANs):** The primary objective of generative models is to evaluate and learn the probability distribution of a training data set. Based on the estimated probability distribution, GANs can generate more data samples to augment the training data set [120]. The GAN framework is made up of two neural networks namely, a generative network (generator) and a discriminative network (discriminator). The generator attempts to capture the distribution of real data and produces new data identical to the real data, while the discriminator tries to differentiate between the real and generated data [121]. Fig. 9 depicts the schematic architecture and computation procedure of GAN for generating state-action value. The generator G and the discriminator D are fed with inputs which are random variables (s', a') and real data (s, a) , respectively. In the context of EH IoT, (s', a') can be regarded as the random state variable of an EH node such as harvested energy and the selected action e.g. transmit power. If the generated samples from G i.e., $G(s', a')$ is taken as

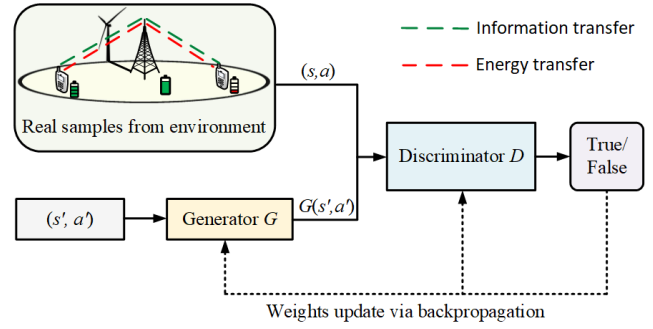


FIGURE 9. GAN architecture for generating state-action value.

an input by the D , being a binary classifier, D should label it with 0, indicating the probability of $G(s', a')$ being a fake sample. The opposite is the case when (s, a) is taken as an input by the D . The D 's goal is to achieve correct classification of the input data whereas G 's purpose is to confuse D to a maximal extent. This adversarial optimization process is repeated until D is unable to differentiate between the real and fake data [122], [123], [124]. A notable application of GAN in EH IoT networks to complement a DRL solution is demonstrated in [125]. In the study, the GAN aims to learn the probability distribution of the state-action value based on the interaction of a legitimate-jammer pair in the network. The SINR, channel coefficients, and battery capacity of the legitimate and jammer nodes are denoted as the state elements while the selected transmit time duration and transmit power represent the actions taken by the nodes. The goal is to learn the optimal state-action value that minimizes the secrecy outage probability (SOP) in the network. The result obtained shows a 67% rise in SOP in the presence of ten eavesdroppers which is the same as the SOP value obtained in the case where one eavesdropper without GAN is considered.

C. DEEP REINFORCEMENT LEARNING

A typical reinforcement learning agent gains the entire knowledge of the system by exploring it to optimize its policy. However, in a large-scale IoT network scenario, this strategy may yield intractable solutions due to the curse of dimensionality. This prompted the combination of deep learning with reinforcement learning, referred to as deep reinforcement learning (DRL). The deep learning leverages its powerful function approximation feature to approximate components of RL such as value functions (e.g Q -functions) or policy functions (e.g policy gradients), or both, with low-dimensional representation [126]. We present notable contributions under the following headings:

1) VALUE-BASED DRL

A distinctive representative of value-based DRL is the deep- Q network (DQN), which uses the CNN to approximate the optimal action value function [127]. The DQN addresses

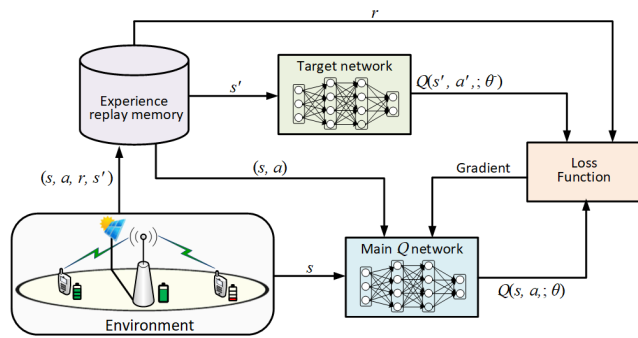


FIGURE 10. DQN structure.

the instability and divergent issues associated with RL when a non-linear function approximator, like neural networks, is used for value function approximation. It achieves this by using an experience replay mechanism and a fixed target network. In experience replay, the DQN stores its observation sequence (s_t, a_t, r_t, s') in the replay memory. Moreover, the sequence is sampled randomly to realize uncorrelated data and to mitigate variations in data distribution changes. Minibatches from the replay buffer are retrieved randomly to train the neural network. A target network maintains its separate parameters and updates periodically the main Q -network values. Thus, the correlations between the estimated Q values and the target $r + \gamma \max_{a'} Q(s', a'; \theta^-)$ are reduced. This process is illustrated in Fig. 10. Based on Bellman's equation, the DQN minimizes the loss function $\mathcal{L}(\theta)$ as

$$\mathcal{L}(\theta) = \mathbb{E} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a, \theta) \right)^2 \right], \quad (12)$$

where θ^- and θ , are the parameters of the target network and Q -network, respectively.

Applications: Application of DQN in EH IoT networks has been demonstrated in [128], [129], [130], [131], [132], [133], [134], [135], [136], [137], and [138] as summarized in Table 9. In [128], the impact of selected modulation level and transmit power by an EH device on the throughput of the system is investigated. The goal is to develop an optimal transmission policy with respect to the energy arrival, battery level, and channel coefficient. A faster convergence is achieved under different bit error rate (BER) settings compared to the traditional Q-learning. In [129], the authors improve the throughput and energy-saving capacity of EH devices by deriving the optimal contention windows in the network. Findings show that at 150 epochs, the realized throughput outperforms the benchmark approaches with different window sizes. In [130], the authors develop a link selection policy for throughput maximization in a cooperative network. Moreover, the DQN is pretrained with data from an MDP solver to accelerate the training convergence of the DQN. In [131], the EH maximization problem is formulated under imperfect CSI condition. Beamforming vectors optimization with a secrecy rate constraint is proposed to solve the problem. In [132], the authors highlight the need

for optimized network topology to enhance the optimal application routing policy and energy allocation for different runtime tasks. Based on the proposed network topology, a joint routing and energy allocation policy is developed with the goal of maximizing the information delivery rate to the destination node. Results obtained outperformed the geographic-based routing without DL and the closed-form solution.

Aside from deep ANN employed in previous studies, other types of DNNs have been proposed to approximate Q -functions. In [133], a CNN-based DQN is proposed to determine an optimal offloading policy in a MEC-based network. To achieve this, the EH device selects the MEC server for offloading and the rate of computational tasks to be offloaded based on its battery level, harvested energy, and radio link data rate. Moreover, the device analyzes its reward, which is formulated as a function of energy consumption, data sharing gain, task drop loss, and computational latency, to improve its policy. The application of LSTM cells has also been considered for improving DQN performance. This concept is demonstrated in [134] for approximating the Q -values associated with the channel and battery states of EH devices. The derived values are used to develop a channel access control policy that maximizes the system sum-rate in the uplink. With fewer training steps, results obtained outperform the data-driven and myopic baseline methods. In another related study [135], the data queuing level is introduced in addition to the battery level in the state space. Moreover, the states are considered partially observable. The goal is to derive an optimal uplink transmission power control policy that minimizes the data queuing delay and battery depreciation cost in a distributed fashion. In [136], the authors develop an optimal EH and packet transmission scheduling that minimizes the average weighted sum-AoI. Aside from approaching the same convergence as the value-iteration approach, additional findings show that the AoI monotonically decreases with decreasing update packet size, which translates to an increase in energy savings to support future transmission.

To avoid the problem of overestimation, in the DQN network, the idea of double DQN (DDQN) is proposed, where action selection is decoupled from the action evaluation [139]. In the context of the EH IoT network, this concept is demonstrated in [137]. In the study, the AoI minimization problem in cooperative communication system is investigated. To solve the problem, a relay selection policy that minimizes the AoI based on the source-relay or relay-destination links is developed. Moreover, unlike the uniformly sampled experience adopted in previous studies, a prioritized experience relay based on the magnitude of the loss function is proposed to improve the learning efficiency of the network. Findings show that a higher number of selected relays and a larger energy buffer minimize the AoI, with the proposed solution having the overall best performance. Another noble application of distributed DDQN is demonstrated in [138] where the tradeoff analysis between the

TABLE 9. Summary of DQN applications in EH IoT networks.

Ref	Network scenario	EH Type	Main Objective	Input	Specific Method	Output	Main Findings	Remark/Limitation
[128]	Wireless network	Solar, RF and wind	Optimal transmission policy for throughput maximization	Channel gain, battery level and harvested energy	DQN	Estimated transmission power	Higher throughput with fewer iterations under different BER and harvested energy is achieved compared to Q-learning.	The study is limited to a SISO link, which may not reflect the robustness of the solution in networks with multiple point-to-point links.
[129]	Wireless local area network	Not specified	Optimal backoff window size for throughput maximization and energy outage minimization.	Channel coefficient, harvested energy, and battery level	DQN	Estimated backoff window size	Findings show that at 150 epochs, the realized throughput outperforms the benchmark approaches with different window sizes.	Benchmarking the solutions with the results from the model-based RL techniques can provide further clarity on the quality of the experiments
[130]	Cooperative communication network	Not specified	Link selection policy for throughput maximization	Channel coefficients, data and energy buffer level	DQN	Optimized link selection	Higher average reward with faster convergence rate is observed compared to the robin round and greedy-based methods.	The computational complexity of the algorithm may be high due to the pretraining module incorporated in the deep Q network.
[131]	Wireless powered network	RF	EH maximization	Channel gains	DQN	Optimized beamforming vectors	In terms of harvested energy, energy efficiency and secrecy capacity, the DQN provides the best performance in comparison to DDQN, DNN, and SCA methods.	Additional practical inputs to the DQN network such as the battery levels and harvested energy of EH devices, could improve the practicality of the solution.
[132]	Wireless sensor network	Solar and wind	Routing and energy allocation policy for data delivery rate maximization	Data packet, energy level, and queue size.	DQN	Optimized route selection and allocated energy.	Achieves better performance in comparison with the Q-learning method, geographic-based routing without DL, and the closed-form approach.	Incorporating energy management schemes in the design, such as duty cycle optimization can enhance the system energy efficiency.
[133]	MEC-based network	RF, solar and wind	Offloading scheme for system utility maximization	Battery level, transmission rate, and harvested energy	CNN-based DQN	Optimized offloading rate	Significant improvement in terms of energy consumption, data sharing gain, task drop loss, and computational latency is achieved compared to Q-learning.	Only local and edge offloading processes are captured. The inclusion of cloud offloading as an alternative is needed for a realistic network design.
[134]	Wireless network	Wind, solar, hydro	Joint battery prediction and access control for sum-rate maximization.	Channel state and battery level	LSTM-based DQN	High battery prediction accuracy and optimized user scheduling	The average sum-rate and the battery prediction loss approaches the solution where non-causal knowledge of the system is assumed.	The energy arrival rate, which is critical to developing a realistic state model of and EH device, is considered.
[135]	Wireless network	Not specified	Uplink power control for minimizing data queuing delay and battery depreciation cost.	Battery level, harvested energy and consumption rate	LSTM-based DQN	Estimated uplink power allocation	The cost function converges toward the baseline DDPG method that relies on the central control unit.	Relevant causal parameters such as channel fading and energy arrival rate are not considered in the state space framework
[136]	Wireless sensor network	RF	Joint EH and packet transmission scheduling for AoI minimization	Battery level, observed AoI and channel gains	LSTM-based DQN	Allocated harvested energy and scheduled packet	The convergence rate compares favourably with baseline study where a value iteration algorithm is adopted.	The solution tends to incur higher computational complexity due to the additional storage unit for experience replay.

TABLE 9. (Continued.) Summary of DQN applications in EH IoT networks.

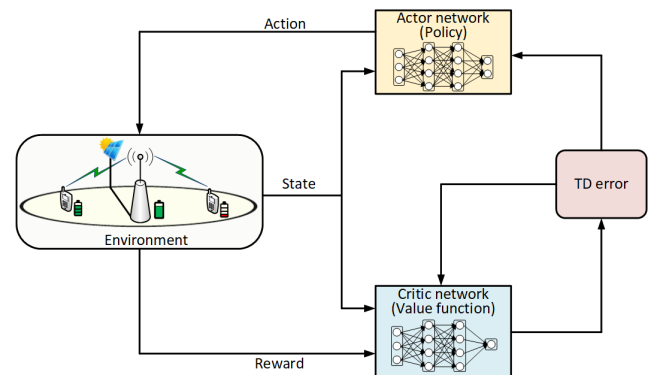
Ref	Network scenario	EH Type	Main Objective	Input	Specific Method	Output	Main Findings	Remark/Limitation
[137]	Cooperative network	RF	Optimal relay selection for AoI minimization	Channel gains, data and energy buffer levels	DDQN	Optimized number of selected relays	Findings show that a higher number of selected relays and a larger energy buffer minimize the AoI with the proposed solution having the overall best performance.	The study relies on the quality of source-relay or relay-destination links for relay selection. However, using the average SNR on both links can provide better performance.
[138]	Wireless networks	RF	Beamforming and power splitting ratio optimization for spectral efficiency and EH tradeoff analysis	Link interference power, data rate, and channel gains	DDQN	Near-optimal estimate of beamforming and power splitting ratio	A tradeoff performance of 1.23 times of the result obtained under a more sophisticated Advantage Actor-Critic is achieved	The solution can be further improved by incorporating more practical state parameters such as battery level and capacity and channel gains.

spectral efficiency and EH is carried out. In the study, the transmitter executes the DDQN algorithm to determine the selection of optimal beamforming pattern and power splitting ratio by its associated receivers. Results obtained slightly outperformed the case where a more sophisticated actor-critic algorithm is employed.

2) POLICY-BASED DRL

In problems with continuous or infinite action spaces, learning a policy function is deemed more efficient in terms of training convergence than value functions. The policy-based DRL chooses a parameterized policy and uses either gradient-based or gradient-free optimization methods to update the parameters [140]. The policy gradient methods are more sample-efficient when numerous parameters are included in the policy; hence, they have been widely adopted in DRL solutions. REINFORCE, Trust region policy optimization (TRPO), and, Proximal Policy Optimization (PPO) are popular examples of gradient-based methods.

Applications: The sole application of policy gradient methods has yet to receive significant attention in the context of EH IoT networks. This may be attributed to their slow convergence rate because the computation relies on episodic tasks to update its parameters consequently, leading to a large variance in gradient estimation [141]. However, considerable studies, as presented in Section III-C3, have combined the policy gradient DRL methods with value-based DRL methods to alleviate this drawback. Nonetheless, a notable attempt to demonstrate the application of the policy gradient approach is provided in [142]. In the study, an hierarchical algorithm is proposed to balance the AoI-energy tradeoff problem caused by the user scheduling in an IRS-aided RF-powered network. The outer loop of the algorithm uses the PPO algorithm to achieve an optimal user scheduling policy. To further reduce the state space in the inner loop, an alternating optimization algorithm is proposed for downlink energy transfer and uplink information transmission. The PPO-based algorithm outperforms its DQN counterpart in terms of reward and convergence rate.

**FIGURE 11. Actor-Critic structure.**

3) ACTOR-CRITIC METHODS

The actor-critic methods are developed from the combination of value-based and policy gradient methods, as depicted in Fig. 11. The actor network learns a policy by using the feedback, i.e., the value function, received from the critic network, and uses the value function for bootstrapping. Thus, this approach tradeoff the variance reduction of policy gradients with the introduction of overestimation bias [61], [143]. Further, it is noteworthy that the actor-critic methods can be categorized based on the type of policy gradient methods adopted, which may be deterministic or stochastic. A popular actor-critic method for deterministic policy gradient is the deep deterministic policy gradient (DDPG) method. The soft actor-critic (SAC) and asynchronous advantage actor-critic (A3C) algorithms are notable actor-critic methods for stochastic policy gradient [144].

Applications: The application of DDPG and its variants have been widely investigated to proffer several solutions in the context of EH IoT networks [145], [146], [147], [148], [149], [150], [151], [152], [153], [154], [155], [156], [157], [158], [159]. A summary of these investigations is presented in Table 10. In [145], the authors develop an energy management policy that maximizes the average net bit rate in point-to-point and one-way relay networks.

In the study, the actor network is trained from a state space comprising the battery level, instantaneous channel power, and harvested energy. Further, the concept of state normalization is introduced to accelerate the training process. In [146], the study leverages the correlation in the information collected by the battery-powered and EH sensors to develop an information update scheduling policy. The goal is to maximize the lifetime of battery-powered sensors while minimizing the AoI in the system. A 15% increase in the network lifetime is achieved compared with closed-form optimization methods. Other notable areas of application of DDPG include: harvested power and time allocation for throughput maximization [147], joint transmit power and energy harvesting control for energy cost minimization [148], joint EH-spectrum access mode selection and power allocation for throughput maximization [149], and joint transmit power, beamforming vector and power splitting ratio optimization to balance the tradeoff between the achievable data rate and harvested energy [150].

Other notable improvements to the DDPG application have been proposed. These include the use of KNN to discretize the continuous action space to improve the training convergence rate of the network [151]. Nonetheless, this approach may increase the computational overhead. Interestingly, recent studies reveal that both continuous and discrete action spaces can coexist. This scenario is investigated in [152] and [153]. The study in [152] demonstrated the joint minimization of device offloading execution time and energy consumption. In the study, the local computation capacity and offloading ratio (the output of the actor network) and server selection (the output of the critic network) are regarded as continuous and discrete actions, respectively. In [153], a long-term capacity maximization problem is investigated. This problem is addressed from the viewpoint of joint relay selection (considered a discrete action) and continuous power allocation optimization problems which are solved by the DQN and DDPG, respectively. In [154], the authors introduce the concept of an action adjuster to prolong the battery life while improving the learning efficiency of the DDPG algorithm. Modeled as a function of the secondary user transmit power and time allocation ratio and the corresponding achievable rate, the action adjuster serves as a decision threshold for activating the actor and the critic networks. A faster convergence and reward of 30% higher than the conventional DDPG is obtained. Additional modifications to improve the performance of DDPG based on the incorporation of LSTM unit are demonstrated in [155]. The study aims to maximize the uplink average sum-rate while minimizing battery prediction loss. Formulated as a two-stage problem, in the first stage, the battery prediction based on the battery and channel states is estimated in the LSTM unit. In the second stage, the power control policy is derived using the actor-critic unit. In addition to the lesser training time, the solution achieves a sum-rate of approximately 24%, higher than the DQN method. Other variants of DDPG have been proposed to address

the problem of overestimation bias. This includes softmax deep double deterministic policy gradients (SD3) [156], double centralized critics [157], and twin-delayed DDPG (TD3) [158], [159]. In [157], the authors derived a joint resource allocation and task offloading policy to minimize the energy consumption of EH devices in a MEC-enabled NOMA system. Moreover, critic regularization and soft value estimation methods are incorporated into the study to accelerate the training convergence. Results obtained indicate better convergence in comparison with DQN and the conventional DDPG. In [156], the transmit power, phase shifters, and scheduling time are jointly optimized to solve the EH and ID rate maximization problem simultaneously in the time and space domain. In [158], the optimal offloading policy to an edge server is derived as a function of battery level, task completion and latency. In [159], the joint time allocation for EH and power allocation policy to maximize the sum-rate of the UE is investigated. The solution achieved a higher sum-rate at approximately 30 episodes compared with the traditional DDPG and the experience-relay-based DDPG.

IV. LESSONS LEARNED

- Selecting suitable ML algorithm:** Choosing an appropriate ML algorithm to optimize an objective function in an EH IoT network depends on several factors. Among the crucial ones is the knowledge of network parameters such as energy and data arrival rate, channel coefficient, or knowledge of statistical information such as the state transition probabilities. In addition, data size is another key factor that cannot be ignored since IoT devices are envisaged to be deployed on a large scale. However, the EH IoT network is stochastic due to unpredictable energy sources and fading wireless channels. Moreover, for data-driven algorithms such as supervised learning, the cost of data acquisition in an unpredictable network environment may be expensive or infeasible. Due to the aforementioned factor, the model-free RL algorithms such as Q -learning and SARSA are deemed suitable for applications in EH IoT networks. Nonetheless, Q -learning and SARSA exhibit poor learning rates under a large state-action space. Though the concept of after-state has been demonstrated in a few studies to address this problem [70], [71], [87], the complexity of this approach scales with the increasing number of state-action pairs. Interestingly, the combination of DL and RL algorithms, known as DRL, has proven to significantly tackle the aforementioned problems. In this regard, the DL is employed to approximate the components of RL, such as the value function or policy function.
- Analytical framework for RL problem formulation:** The frequently adopted mathematical framework for modeling RL problems in EH IoT networks is the MDP. The MDP is state-dependent and provides sequential decision-making problems based on the assumption that the perfect knowledge of the state. However, this assumption may be invalid in scenarios where only partial knowledge of the state is available. Hence, POMDP is developed

TABLE 10. Summary of actor-critic DRL applications in EH IoT networks.

Ref	Network scenario	EH Type	Main Objective	Input	Specific Method	Output	Main Findings	Remark/Limitation
[145]	Wireless sensor network	Solar	Energy management policy optimization average net bit rate maximization	Battery level, channel gain, and harvested energy	DDPG	Normal-ized values of parameters	Compared with Lyapunov optimization and Q-learning methods, the average bit rate increases by four and ten times, respectively.	Extending the study to a multi-user network is important to further examine the effectiveness of the proposed algorithm.
[146]	Wireless sensor network	Solar	Information update scheduling policy network lifetime maximization	Battery level and harvested energy.	DDPG	Estimated transmission status update	A 15% increase in the network lifetime is achieved with the proposed method compared with the closed-form methods	Power allocation and state normalization can enhance the energy management and training convergence of the system, respectively.
[147]	Wireless sensor network	Solar and wind	Optimal power and time allocation policy for throughput maximization.	Battery level, harvested energy and channel coefficients	DDPG	Optimized transmit power and time allocation	Results obtained outperform the DQN due to its robustness to quantization error	Though faster convergence is achieved, higher computational complexity remains the main challenge.
[148]	UAV network	RF	transmit power and energy harvesting control for system energy minimization.	Sensed data and battery level	DDPG	Optimized time slot and power allocation	A lower energy consumption cost is achieved compared with the greedy-based and no-energy-control algorithms.	The solution can be further improved by incorporating routing dynamics in the state-action space.
[150]	Wireless powered network	RF	joint transmit power, beamforming vector and power splitting ratio optimization for data rate maximization	Power splitting ratio, beamforming vector, transmit power	DDPG	Optimized values of input parameter	The DDPG algorithm outperforms its DQN counterpart indicating its capacity to handle the continuous action space effectively	Incorporating of additional state elements such as battery level, harvested energy, data buffer level can further improve the practical realization of the proposed solution
[149]	Cognitive radio network	RF	Joint EH-spectrum access mode selection and power allocation optimization for throughput maximization	Channel gain and battery level	DDPG	Estimated power allocation and channel gains	The DDPG achieves an average throughput of 20% higher than its DQN counterpart.	The system performance is centered on the secondary UE throughput with little reference to the performance of primary UE.
[151]	Wireless heterogeneous network	RF	Energy efficiency maximization	Battery level of SBS and node locations	Wolpertinger DDPG	Optimized EH and data transmission time	At less than 10 episodes, the proposed solution outperforms the ones based on the traditional DDPG and DQN	The performance may come at the cost of higher computational complexity due to the inclusion of KNN unit in the framework.
[152]	MEC-based network	Not specified	Server selection, continuous offloading ratio, and local computational capacity for execution time and energy consumption minimization	Size of computation task, channel gain, harvested energy etc.	Hybrid DDPG-DNN	Optimized number of selected servers	Slower convergence of the reward function is observed under a multiuser scenario due to complex offloading decisions compared with a single user scenario.	Investigating the problem under strictly DDPG and DNN is needed to highlight the efficiencies of the proposed hybrid DDPG-DNN schemes
[153]	Underwater cooperative communication	Not specified	Joint power allocation and relay selection for capacity maximization	battery level and channel gains	Hybrid DQN-DDPG	Estimated channel gains and allocated transmit power	A faster convergence is observed compared with the baseline DQN	Main challenge with this approach lies in its higher computational complexity which increases with network scalability

TABLE 10. (Continued.) Summary of actor-critic DRL applications in EH IoT networks.

[154]	Cognitive radio network	RF	Joint time allocation, transmit power control, and spectrum access optimization for packet loss minimization	Channel gains harvested energy and battery level	Action adjuster-based DDPG	Estimated packet transmission throughput	Compared with the conventional DDPG, a 30% increase in the reward with faster convergence is achieved.	The introduction of the action adjuster network may increase the computational overhead of the solution.
[155]	Wireless communication network	Solar	Access and power control policy for average sum-rate maximization	Battery level, channel coefficients, and harvested energy	LSTM-DDPG	Optimized allocated power	Results show that fewer training steps are required to achieve results of 24% higher than DQN approach	The training convergence can be further enhanced by the incorporation of state normalization.
[156]	UAV-IRS network	RF	Joint optimization of transmit power, phase shifters, and scheduling time for EH maximization	Channel gains, node locations and antenna position	SD3	Optimized values of transmit time and power allocation and phase shift	An average harvested energy of 64.2% is achieved, which approaches the upper limit, calculated at 67.6%	Details on user association with the UAV are not captured, which is a major requirement for realistic UAV-aided network.
[157]	MEC-based network	RF	Joint resource allocation and task offloading offloading policy energy consumption minimization	Battery level, queue length, channel gains, etc.	Double centralized critics	Estimated EH duration, offloading ratio, and transmit power allocation	Achieves better performance and faster convergence in comparison with DQN and the conventional DDPG.	The design is based on the assumption of perfect successive interference cancellation which may not be realistic in practice.
[158]	MEC-based NOMA network	RF	Optimal offloading policy with energy-saving, latency and task completion constraint	Data size and channel gains	TD3	Optimized offloading policy	A latency less than 47%, task completion rate greater than 20% and 21% improvement in energy saving is achieved in comparison with DDPG	The energy-saving of the system can be improved by incorporating a decision maker that switches between local computation on the UE terminal and edge computation
[159]	Cognitive radio network	RF	EH time and power allocation policy for sum-rate maximization	Battery level	TD3	Estimated time slot for EH and data transmission	Achieve a higher sum-rate compared with conventional DDPG and also require 30 episodic tasks to converge.	Incorporation of state normalization and LSTM unit can improve the convergence rate of the solution.

to relax the state uncertainty constraints. Though MDP has been widely adopted in most EH IoT RL-based solutions, a notable application of POMDP can be found in [81] where EH source and relay nodes are considered to have partial knowledge of harvested energy, channel gains, data buffer levels, and battery levels. Aside from MDP-based frameworks, another alternative mathematical framework is the mean-field game. Although this approach is rarely used, its application in the context of EH IoT has been demonstrated in [160] to combat the curse of dimensionality in the MDP framework. The goal of the study is to develop a distributed power control policy for sum throughput maximization.

- *Performance of DL algorithms:* The deep ANN, CNN, and LSTM-RNN constitute key DL algorithms adopted in EH IoT networks. However, these algorithms have been used in different contexts. For instance, the deep ANN has been widely employed in regression-based problems such as evaluating the impact of channel gains on transmit power

and power splitting ratio [99], the impact of harvested energy, the number of antennas of EH devices on system throughput, and so on. Nevertheless, the deep ANN is also affected by the “curse of dimensionality” due to the full connectivity of neurons from one layer to the next. The CNN can effectively combat this problem due to its sparse connectivity, equivariant representations, and parameter-sharing features. Since CNN is usually employed in classification-based problems, its application in EH IoT has yet to receive significant attention. Nonetheless, a few investigations have employed it for channel estimation [105], and outage probability prediction [108]. The LSTM-RNN is another important DL algorithm that has been employed to reduce the state space dimension, but its sole application in EH IoT networks is rare. Nonetheless, various attempts have considered its application in the DRL domain for battery prediction [134], [155], relaxation of partially observable state constraints [135], and EH and transmission scheduling [136],.

- *Training convergence*: It is usually desired to achieve fast convergence during the training phase of a DL model. However, this may be challenging under a large-scale network as they are characterized by high-dimensional state space. A key strategy to accelerate training convergence is to use the pretraining approach. Here, the output data from traditional optimization methods such as iterative methods or closed-form solutions are used for pretraining purposes. This idea is demonstrated in [102], [103], and [130], where EH data from sequential parametric convex approximation method, Lagrangian iterative method, and MDP solver are employed to pretrain DQN network, respectively. In addition, it is shown that some training algorithms could potentially accelerate training convergence. A notable example of such is the ADAM optimizer demonstrated in [101] and [110].
 - *Training failure*: Among the prominent causes of training failure or poor performance of a learning algorithm is the problem of overestimation bias, as in the case of TD learning, vanishing and exploding gradient, overfitting, and large variance, as in the case of DL and DRL. To tackle these problems in EH IoT networks, the concept of a double Q network is demonstrated in [137], to combat the overestimation bias. The activation functions such as ReLU and ELU have been frequently adopted to tackle the problem of vanishing gradients [99], [100], [101], [102], [103]. Since these problems also impact value-based DRL and policy-based DRL, the development of hybrid value-policy DRL, known as the actor-critic method, is proposed to balance the tradeoff between the overestimation bias and large variance, which are peculiar limitations of the value-based DRL and policy-based DRL, respectively. In the context of EH IoT networks, the DDPG and its modifications, as highlighted in Table 10, have been widely employed.
 - *Optimizing EH ML model*: Aside from developing an optimal policy to improve the accuracy and performance of an ML model in EH network, ensuring their robustness to training data variability in a real-world environment is important. To achieve this, some studies have demonstrated the use of regularization methods such as the dropout method to combat the problem of high variance in test error [102]. Further, modifying the network infrastructures can bring about lower computational complexity during data acquisition for the network and training of the ML model. For instance, some studies have incorporated MEC strategy to optimize the computational tasks [87], [88], [133], [152], [157], [158]. This approach not only improves latency and bandwidth efficiency of the system but also reduces the energy cost associated with the transmission overhead.
- data acquisition can be achieved from the outputs of the traditional closed-form or iterative approaches. To accommodate the uncertainties associated with the EH environment, several experiments are expected to be conducted under different scenarios and optimization constraints to improve data quality. Since a model performs better with large training data, this may lead to an expensive cost of data acquisition, consequently increasing the implementation cost.
- *Hardware design*: In situations where miniature EH IoT devices are required, such as medical implants or covert surveillance cameras, integrating ML processors while attempting to realize compact electronic hardware that can support the complex computational task may be difficult. Notable key factors to this are the energy consumption and heat generation associated with intensive computational tasks. Since EH sources are intermittent, a viable means to address this challenge is to increase the storage capacity. Moreover, a cooling unit may be required to dissipate the heat generated by the processor. Consequently, developing miniaturized hardware to accommodate these features may be challenging.
 - *Real-time application*: Due to the unstable nature of EH sources, various energy-efficient solutions are usually incorporated in the design objective. Among the prominent energy-saving solutions is the duty cycle-based power consumption management approach. To develop a robust solution, the sleep and awake operation of the EH device are designed around several factors, such as the state of battery level, weather, harvested energy, data arrival rates, etc. However, going through these processes may increase the latency of the system. Also, in complex systems with billions of EH IoT devices, the data processing is best handled via a centralized system with a stronger computational power. Nonetheless, the process of data upload and downloading will compound the transmission delay in the system. Consequently, this may result in poor performance in scenarios where real-time processing is required. For instance, some EH Internet of medical things (IoMT) are required for real-time monitoring of patient health for emergency response.
 - *Environmental constraints*: The operation of EH IoT devices in harsh or delicate environments can be challenging. For example, in the case of EH IoMT devices, in order not to violate the maximum permissible exposure of human tissue to electromagnetic radiations, the RF transmission must be kept below a certain level [161]. Hence, this could affect the EH efficiency since the available energy is bounded by the allowable RF transmit power. In the case of nanogenerators, the harvesters require mechanical activities of the human body to create charge excitation in the nanomaterial. However, it is unrealistic for the human body to be in perpetual motion. Hence, these limitations could affect the data quality and accuracy of ML models.

V. CHALLENGES

- *Implementation cost*: The performance of DL and DRL in EH IoT largely depends on the volume and quality of training data. It has been demonstrated that training

- *Reliability and Maintenance*: The purpose of EH is to minimize reliance on energy buffers. However, these buffers are still required as a backup since green energy sources are stochastic. Thus, in a large-scale network, the cost of maintenance, such as battery charging or replacement, remains an open issue. From the viewpoint of ML solutions, the use of empirical data for training purposes and regular program updates are expected due to improvements in ML algorithms. However, due to data privacy and other policies to enhance the system security. Some of these updates may require manual approval from the user end. This as well may become a difficult task in a large-scale network.
- *Security*: The quality of data generated from EH devices may be compromised due to variations in energy arrival rate. This may result in poor training performance of an ML model. From the viewpoint of wireless data transmission, the collection of training data may be prone to an adversarial attack due to the openness of the wireless channel. Thus, sophisticated encryption algorithms are required to ensure data protection. However, aside from an increase in energy consumption, incorporating complex encryption programs may also contribute to an increase in implementation cost.

VI. OPEN AREAS

In the following, we highlight open areas that can be explored for further study.

- *Development of a lightweight ML algorithm*: From the literature, it is evident that the application of DRL in EH IoT networks has caught the attention of the research community. However, the complexities of DRL algorithms scale with the size of the network, and a sophisticated algorithm may be required to alleviate this problem. This may be cost-ineffective, and also, the energy consumption of the device may increase due to the associated complexity of the ML model. Hence, developing a lightweight algorithm while preserving its performance is crucial. A viable means of achieving this is via the incorporation of model compression techniques in the design framework. Compression techniques such as the pruning method, quantization approach, low-rank factorization, etc. [162], can be explored.
- *Exploring efficient training data generation strategy*: Employing the output data from classical optimization approaches has been widely explored as a means for generation of training data. However, the output data from these studies are largely based on simulations which could pose a result validation problem in a real scenario. A viable solution to this challenge is to employ an empirical data generation approach. This strategy has been successfully demonstrated in [163], for data generation in wireless sensor networks.
- *Hardware solution*: For seamless IoT device mobility, the use of solar cells or rectenna for EH is deemed practical. However, for efficient EH, a larger surface area may be required in the case of solar cells, which may lead to an undesired increase in the size of the EH device. Due to this factor, and because RF signals are more predictable, EH from the RF signals is considered more advantageous in terms of energy supply predictability and hardware complexity since less-complex modification to the receiver antenna is required [164], [165], [166]. However, the efficiency of harvested RF energy is constrained by critical factors such as the distance from the RF source to the EH device, the channel conditions, the number of rectennas, etc. A viable approach to address this issue is to explore multi-band RF antennas that can scavenge energy from ambient RF signals. Moreover, this approach can be enhanced by the incorporation of ML algorithms for channel estimation and reliability prediction purposes.
- *Exploring of generative DL model*: In the DL category the discriminative models such as deep ANN, CNN, and RNN have been widely employed to solve various optimization problems in the EH IoT network. However, most of these models fall under the domain of supervised learning; consequently, data labeling is a major challenge in adopting this approach. Alternatively, generative (unsupervised) learning models suitable for handling high-dimensional data, such as AEs, GANs, and RBM can be adopted. Though AE and GAN applications have been demonstrated in [119] and [125], respectively, there exist other variants of the aforementioned unsupervised DL that have yet to be explored.
- *ML in EH visible light communication (VLC) IoT networks*: In recent years, studies on EH from solar and lightwave for powering IoT devices in VLC systems have become a trending topic. However, the application of ML in this area has yet to gain the attention of the research community. Some of the exciting areas application of that can be explored include data and energy arrival prediction to aid various optimization solutions, energy management for improved energy efficiency, data outage probability prediction, and so on.
- *ML in EH IoUT*: From an underwater perspective, there has been a dramatic rise in the deployment of IoUT nodes for various purposes such as surveillance, scientific research and exploration, commercial and so on [167], [168]. As a result, the application of EH technologies in underwater communication systems has received significant attention for the sustainability of IoUT networks. Surprisingly, a few studies such as [59] and [153] have proposed the use of MAB and DRL algorithms for throughput and capacity maximization, respectively. Hence, significant ML-based research is still required in this area of study.
- *Security and Privacy*: Adversarial attacks in EH IoT networks can emerge in various forms. These include intercepting and corrupting training data during the

transmission, injection of malicious updates in the network, and various forms of denial of service (DoS) attacks. Various physical layer security approaches, such as cooperative jamming and artificial noise, and lightweight cryptographic techniques have been developed in the literature to combat most of the threats. However, their implementation in EH IoT networks based on ML algorithms has yet to receive the desired attention.

VII. CURRENT TRENDS AND RESEARCH ON EH IOT NETWORKS

Though this paper focuses on applications of ML in EH IoT networks, it is pertinent to also highlight notable trends and current research on EH IoT networks.

- *EH processes*: The EH processes in IoT networks can be grouped into four categories, namely, energy resource, energy conversion, energy storage, and energy consumption [169]. Interestingly, each of the aforementioned categories has recorded notable emerging research worth highlighting. In the energy resource category, popular green energy sources that have been widely incorporated in EH IoT systems include RF, solar radiance, thermal, mechanical (e.g., vibration and motion), and so on [170]. However, the dramatic rise in the development of various wearable devices and the exploitation of biomechanical activities from humans, have led to a surge in the application of nanogenerators such as TENG and PENG [171]. These nanogenerators not only act as power sources but the self-generated electrical signals can be processed for various monitoring functions [172], [173]. In the energy conversion phase, recent focus has been on increasing the harvested energy and the conversion efficiency, since a DC output is required to power the IoT devices. For instance, in RF-based EH, the use of a high-gain multi-band antenna has been employed to increase the EH rate [174]. Also, the use of harmonic termination networks [175], for mitigating dissipation of harvested power has been proposed to enhance the EH conversion efficiency in the rectifying unit. In the case of nanogenerators, to increase the EH performance, an internal resonant design to tune the natural frequencies of the EH circuit has been proposed to adapt to different excitation frequencies [176]. In the energy storage category, research on the development of micro supercapacitors and storage-controlling methods is currently being explored [177], [178]. Finally, in the energy consumption phase, the role of several power management techniques is being investigated to satisfy the power requirement of an IoT device. These include the startup circuit design of the energy harvester, DC-DC conversion design, and incorporation of fuzzy-based controllers to achieve an adaptive operation of the
- entire power management unit [179], [180], [181], [182], [183].
- *EH-aided Federated learning (FL)*: FL is another emerging topic in the ML-aided EH networks. FL can be described as a distributed ML framework developed to tackle the privacy leakage concerns in the traditional distributed ML algorithms while taking into consideration the data and system heterogeneity [184]. Generally, FL-based networks require an iterative process of local computation of model parameters by individual UEs, which are aggregated and refined at a central FL coordinator. The UEs receive the updated global model from the central FL coordinator as a reference for the next iteration phase until the desired accuracy is obtained [185]. However, these computation and transmission processes are energy-intensive, which could lead to quick exhaustion of the IoT batteries, consequently impairing the accuracy of the FL model [186]. Recent developments to address the energy consumption challenge include the incorporation of long-term energy budget constraints in the FL model [187], developing a joint information and energy security framework to combat malicious UEs [188], over-the-air computation approach to minimize the transmission overhead [189], accelerating the convergence rate of the FL model to minimize the system computational time [190], and joint optimization of various system resources such as transmit power allocation [191].
- *EH edge computing-based IoT networks*: IoT devices have limited computational capacity, and to cope with the ever-increasing computational tasks, the concept of cloud computing is introduced [192]. However, cloud computing exhibits its own limitations in terms of a large bandwidth requirement as a result of massive data uploads to cloud servers. Moreover, the process of offloading computational tasks to the cloud servers leads to an inevitable delay in the execution of tasks by the IoT devices [193]. To combat these challenges, MEC strategy is developed, where MEC servers with higher computational capacity than the IoT devices are deployed closer to the IoT device [194], [195]. Interestingly, this strategy has reinforced the research interest in the field of EH IoT networks. For instance, to minimize the energy consumption of IoT devices, some studies have demonstrated the impact of different offloading policies that are constrained by the availability of harvested energy and energy level in the storage device [196], [197], [198]. However, recent studies in the domain of fog computing indicate that additional improvements can be achieved through the incorporation of various ML algorithms. Another trending topic strengthened by the integration of MEC in EH IoT systems is the AoI analysis of the systems. The AoI is a metric that measures the freshness or timeliness of information at a destination node. Recent studies in

this direction have focused on the incorporation of ML algorithms to improve the AoI performance with respect to the harvested energy [88], [136], [137], [142], [146].

- *NTN-aided IoT network*: With the ever-increasing growth of IoT devices, it is projected that the current terrestrial-based communication infrastructures may not support their demands in the foreseeable future [199]. Moreover, establishing wireless services in remote areas such as oceans, deserts, and arctic regions remains a daunting challenge due to several factors associated with power supply availability, network maintenance and upgrade, security of network infrastructure, among others [200]. The emergence of non-terrestrial networks (NTNs) which comprise UAV, high-altitude platforms, and satellite networks, serves as complementary networks to overcome the inherent limitations of the terrestrial networks [201]. Although some recent investigations have been conducted to maximize the performance of IoT devices in NTNs [202], [203], the roles of EH and ML-enabled optimization solutions have yet to be captured. However, these aspects of research cannot be ignored due to their potential role in sustaining the NTNs in the next-generation communication networks.
- *Security and privacy*: The role of data security and privacy cannot be overemphasized in IoT networks. Though federated learning has been identified as one of the key techniques to realize secure data transmission, the malfunction of a central server that aggregates the model parameters may lead to a significant delay [204]. Among the other sophisticated security and privacy protection methods that are gaining popularity in IoT networks is blockchain technology. Blockchain offers robustness against data modification using a distributed ledger technology, making it less susceptible to cyber attack [205]. However, blockchain is not totally immune to malicious attack [206], hence this approach remains an open topic of interest in academia and industry.

VIII. CONCLUSION

In this paper, we have investigated various studies where ML algorithms have been employed for the realization of intelligent EH IoT networks. Due to the stochastic nature of the green energy sources and wireless fading channels, we center our discussion on ML categories widely adopted to accommodate these peculiarities. In this regard, we concentrate on three ML categories, namely reinforcement learning, deep learning, and deep reinforcement learning. Further, we identify and provide an in-depth description of the main algorithms under each ML category. Based on the identified algorithms, we present various articles where their applications are demonstrated. Furthermore, we highlight critical lessons gained from our study and challenges that can impact the implementation of the identified algorithms. Finally, we identify open areas that can be explored for future

research. We are certain that this article will serve as good instructional material for researchers willing to explore the potential of ML algorithms in the EH IoT networks.

As a key insight, considering the unpredictable nature of energy sources and wireless channels coupled with the complexity and cost of training data acquisition, the role of model-free RL models should not be ignored in EH IoT networks. RL accuracy is robust to variability in the environment and also independent of existing data to initialize the learning process since this can be achieved through the interaction of the learning agent with its environment. However, model-free RL algorithms exhibit poor learning rates under a large state-action space; thus, they may be unable to efficiently handle the volume of data generated from billions of EH IoT devices. Interestingly, this limitation can be overcome by leveraging the excellent data processing capability of DL. Therefore, we can infer that DRL will be a key driver to fully support the integration of EH in the 6G and beyond networks. Nonetheless, due to the hardware constraints of EH IoT devices, an energy-efficient, lightweight design of DRL algorithms will be required to minimize the energy consumption associated with their inevitable computational complexity.

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